



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

•Methodology:

- Collected SpaceX launch data via API and Wikipedia scraping
- Cleaned and wrangled datasets for modeling
- Performed EDA using Python visualizations and SQL queries
- Built interactive maps (Folium) and dashboards (Plotly Dash)
- Trained and tuned classification models (Logistic Regression, SVM, Decision Tree, KNN)

•Results:

- KSC LC-39A had highest launch success rate
- ES-L1, GEO, SSO orbits showed near-perfect reliability
- Logistic Regression was the best model (CV: 0.8196, Test: 0.8333)
- Dashboard and maps revealed booster performance and launch site logistics

Introduction

- Project Background:**

An end-to-end data science analysis of SpaceX Falcon 9 launches, combining API data, web scraping, SQL, and machine learning.

- Key Questions:**

- What factors influence launch success?
- Which sites and boosters perform best?
- Can we predict landing outcomes using classification models?

I can also generate a visual overview slide showing these questions mapped to each methodology stage if you'd like.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Queried SpaceX API for structured launch data
 - Scraped Falcon 9 records from Wikipedia snapshot
 - Exported unified datasets for analysis
- Perform data wrangling
 - Cleaned missing values, standardized types
 - Created binary target for landing success
 - Final dataset ready for modeling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Built and tuned multiple classifiers, with Logistic Regression emerging as the best performer (CV = 0.8196, Test = 0.8333).

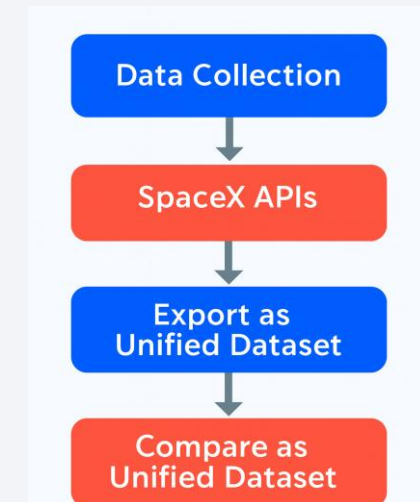
Data Collection

•SpaceX API Integration

- Queried REST endpoints: /rockets, /launchpads, /payloads, /cores using requests.
- Extracted: booster version, payload mass, orbit type, launch site coordinates, landing outcomes, reuse metrics.
- Filtered out Falcon 1 and multi-core/payload entries for consistency.
- Final Output: dataset_part_1.csv
- Static Reference: Used API_call_spacex_api.json for reproducibility.
- GitHub Link -- https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Lab%201_%20Collecting%20the%20data_jupyter-labs-spacex-data-collection-api.ipynb

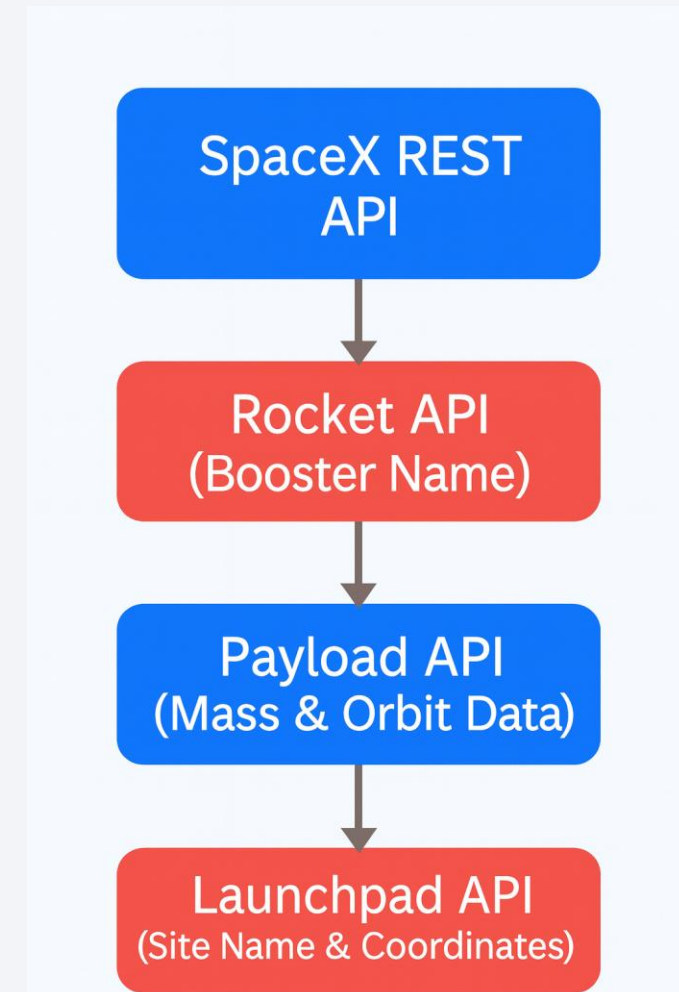
•Web Scraping from Wikipedia

- Source: Static snapshot of Falcon 9 launch records (June 9, 2021).
- Tools: requests, BeautifulSoup, re, unicodedata, pandas.
- Parsed third HTML table → extracted columns: Flight No., Launch Site, Payload, Mass, Orbit, Customer, Outcome.
- Used helper functions to clean nested HTML.
- Final Output: spacex_web_scraped.csv
- GitHub Link -- <https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Web%20scraping%20Falcon%209%20and%20Falcon%20Heavy%20Launches%20Records%20from%20Wikipedia.ipynb>



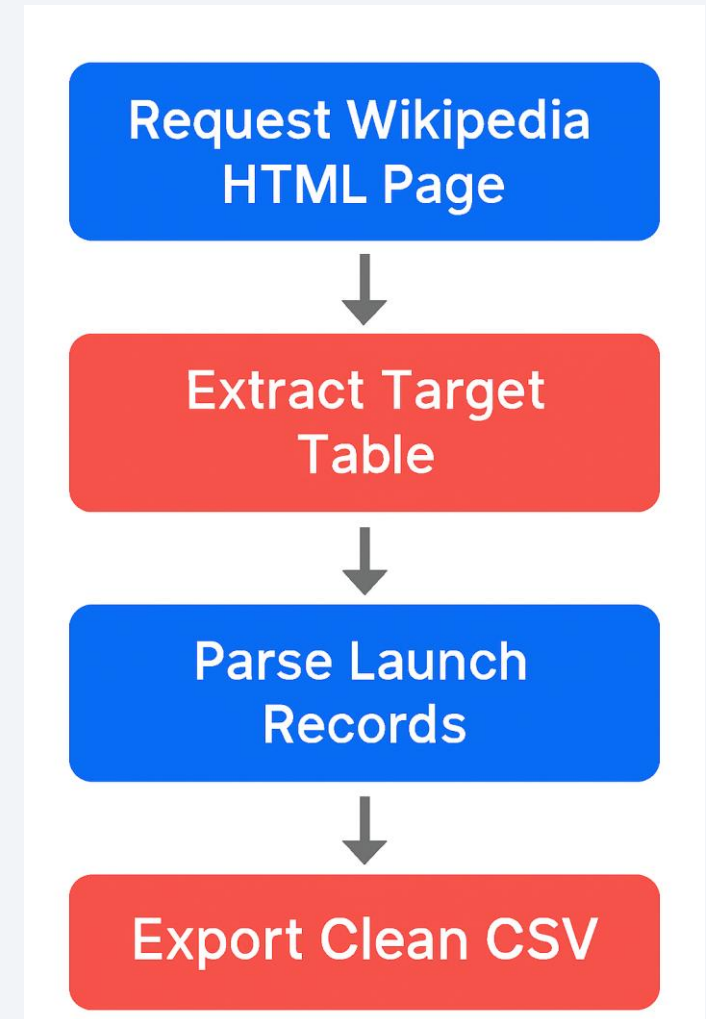
Data Collection – SpaceX API

- **API Source:** Queried SpaceX REST endpoints (/rockets, /launchpads, /payloads, /cores) using requests.
- **Data Extracted:** Booster version, payload mass, orbit type, launch site coordinates, landing outcomes, reuse metrics.
- **Filtering Logic:** Removed Falcon 1 launches and multi-core/payload entries for consistency.
- **Final Output:** Cleaned dataset of Falcon 9 launches exported as dataset_part_1.csv.
- **Static Reference:** Used a static JSON (API_call_spacex_api.json) for reproducibility.
- **GitHub Link**
https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Lab%201%20Collecting%20the%20data_jupyter-labs-spacex-data-collection-api.ipynb



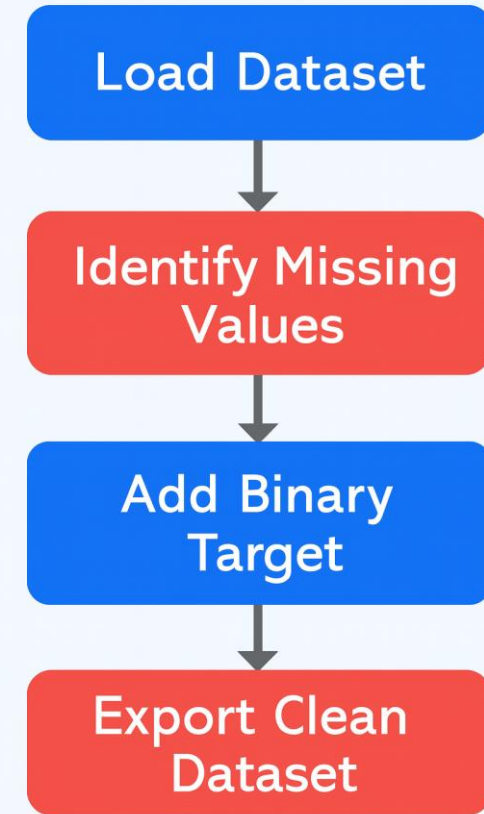
Data Collection - Scraping

- **Source:** Scraped Falcon 9 launch records from a static Wikipedia snapshot (dated June 9, 2021).
- **Tools Used:** requests, BeautifulSoup, re, unicodedata, pandas.
- **Target Table:** Extracted the third HTML table containing launch records using `soup.find_all("table")`.
- **Column Extraction:** Parsed headers to build column list: Flight No., Launch Site, Payload, Payload Mass, Orbit, Customer, Launch Outcome.
- **Custom Parsers:** Defined helper functions to clean and extract nested HTML content (e.g., `date_time`, `booster_version`, `landing_status`, `get_mass`).
- **Data Dictionary:** Constructed `launch_dict` with cleaned values and additional fields (Date, Time, Booster Version, Booster Landing).
- **Final Output:** Converted to Pandas DataFrame and exported as `spacex_web_scraped.csv`.
- **GitHub Link** - <https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Web%20scraping%20Falcon%209%20and%20Falcon%20Heavy%20Launches%20Records%20from%20Wikipedia.ipynb>



Data Wrangling

- **Data Source:** Loaded cleaned Falcon 9 dataset (dataset_part_1.csv) using pandas.
- **Missing Values:**
 - LandingPad: ~29% missing (retained as None)
 - All other columns: 0% missing
- **Data Types Identified:**
 - Numerical: PayloadMass, Longitude, Latitude, FlightNumber, Block, ReusedCount
 - Categorical: BoosterVersion, Orbit, LaunchSite, Outcome, LandingPad, Serial
 - Boolean: GridFins, Reused, Legs
- **Launch Site Distribution:**
 - CCAFS SLC 40: 55 launches
 - KSC LC 39A: 22 launches
 - VAFB SLC 4E: 13 launches
- **Orbit Distribution:**
 - Most common: GTO, ISS, VLEO
 - Rare: ES-L1, GEO, SO, HEO
- **Landing Outcome Classification:**
 - Created binary Class label: 1 = successful landing, 0 = failed/no attempt
 - Success Rate: ~66.7%
- **Final Output:** Exported enhanced dataset as dataset_part_2.csv for modeling and visualization.
- **GitHub Link:** https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Lab%20%20Data%20wrangling_labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

- **Flight Number vs. Launch Site**

- `sns.scatterplot(x='FlightNumber', y='LaunchSite', hue='Class')`
- Purpose: Show launch success trends across sites over time.
- Insight: KSC LC 39A shows higher success; early flights had more failures.

- **Payload Mass vs. Launch Site**

- `sns.scatterplot(x='PayloadMass', y='LaunchSite', hue='Class')`
- Purpose: Compare payload capacity and success across sites.
- Insight: KSC LC 39A handled heavier payloads with mostly successful outcomes.

- **Success Rate vs. Orbit Type**

- `df.groupby('Orbit')['Class'].mean().plot(kind='bar')`
- Purpose: Visualize average success per orbit.
- Insight: ES-L1, GEO, SSO have near-perfect success; GTO and SO show higher failure rates.

- **Flight Number vs. Orbit Type**

- `sns.scatterplot(x='FlightNumber', y='Orbit', hue='Class')`
- Purpose: Track orbit usage and success over time.
- Insight: Later flights trend toward more successful outcomes.

- **Payload Mass vs. Orbit Type**

- `sns.scatterplot(x='PayloadMass', y='Orbit', hue='Class')`
- Purpose: Assess how payload weight affects success across orbits.
- Insight: Heavier payloads succeed more in ES-L1, GEO, SSO; GTO and SO show failures.

- **Launch Success Yearly Trend**

- `df['Year'] = pd.to_datetime(df['Date']).dt.year`
- `df.groupby('Year')['Class'].mean().plot()`
- Purpose: Display success rate evolution from 2010 to 2020.
- Insight: Steady improvement; peak success in 2019 (~90%).

- **GitHub Link** - [https://github.com/ADVAIT135/IBM-DATA-](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/EDA%20with%20Visualization%20Lab.ipynb)

[SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/EDA%20with%20Visualization%20Lab.ipynb](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/EDA%20with%20Visualization%20Lab.ipynb)

EDA with SQL

- Unique Launch Sites

```
SELECT DISTINCT "Launch_Site" FROM spacetable;
```

- Identified all launch locations for filtering and grouping.

- Launch Sites Starting with 'CCA'

```
SELECT * FROM spacetable WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

- Previewed launches from Cape Canaveral-based sites.

- Total Payload Mass by NASA (CRS)

```
SELECT SUM("PAYLOAD_MASS_KG_") FROM spacetable WHERE Customer IS 'NASA (CRS)';
```

- NASA (CRS) missions launched a total of 45,596 kg.

- Average Payload Mass for F9 v1.1

```
SELECT AVG("PAYLOAD_MASS_KG_") FROM spacetable WHERE "Booster_Version" IS 'F9 v1.1';
```

- Mean payload mass: 2,928.4 kg.

- First Successful Ground Pad Landing

```
SELECT MIN(Date) FROM spacetable WHERE "Landing_Outcome" IS 'Success (ground pad)';
```

- Earliest success: 2015-12-22.

- Successful Drone Ship Landings with Medium Payloads

```
SELECT DISTINCT "Booster_Version" FROM spacetable WHERE "Landing_Outcome" IS 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" BETWEEN 4000 AND 6000;
```

- Identified boosters that performed well under specific conditions.

- Count of Success vs. Failure Outcomes

```
SELECT "Landing_Outcome", COUNT(*) AS total FROM spacetable WHERE "Landing_Outcome" IN ('Success', 'Failure') GROUP BY "Landing_Outcome";
```

- Success: 38 launches; Failure: 3 launches.

- Boosters with Maximum Payload

```
SELECT DISTINCT "Booster_Version" FROM spacetable WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM spacetable);
```

- Listed top-performing boosters by payload capacity.

- Failed Drone Ship Landings in 2015

```
SELECT CASE substr("Date", 6, 2) WHEN '01' THEN 'January' ... END AS month_name, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM spacetable WHERE substr("Date", 1, 4) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)';
```

- Revealed seasonal patterns in failed landings.

- Rank Landing Outcomes (2010–2017)

```
SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM spacetable WHERE substr("Date", 1, 10) BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY outcome_count DESC;
```

- Most common: No attempt (10); Drone ship success/failure: 5 each.

GitHub Notebook URL -- [https://github.com/ADVAIT135/IBM-DATA-](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Assignment_%20SQL%20Notebook%20for%20Peer%20Assignment_jupyter-labs-eda-sql-coursera_sqllite.ipynb)

[SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Assignment_%20SQL%20Notebook%20for%20Peer%20Assignment_jupyter-labs-eda-sql-coursera_sqllite.ipynb](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Assignment_%20SQL%20Notebook%20for%20Peer%20Assignment_jupyter-labs-eda-sql-coursera_sqllite.ipynb)

Build an Interactive Map with Folium

- Map Objects Added
- **Markers:**
 - Placed at key launch sites like VAFB (California) and CCAFS SLC-40, KSC LC-39A (Florida).
 - Used to highlight launch pads, support zones, and observation points.
- **Circles:**
 - Represented facility clusters (e.g., 10 in California, 46 in Florida).
 - Used to visualize density and strategic grouping of launch infrastructure.
- **Lines:**
 - Measured straight-line distances from SLC-40 to nearby assets:
 - Airport (~11.2 km)
 - Highway (~8.0 km)
 - City (Titusville, ~22.7 km)
 - Railway (~11.2 km)
 - Used for spatial planning and logistics visualization.
- **Popups & Tooltips:**
- Added to display coordinates, facility names, and strategic notes (e.g., Centaur Road, ECC, NASA Parkway).
- Why These Objects Were Added
- **Operational Context:**
 - Markers and circles show launch site specialization and facility density.
 - Lines and distance annotations support planning for crew transport, emergency access, and cargo logistics.
- **Strategic Insights:**
 - Florida's proximity to equator and infrastructure density supports high-frequency launches.
 - California's polar orbit specialization is visualized through isolated cluster mapping.
- **Public-Private Integration:**
- Visual distinction of SpaceX-operated pads (e.g., 39A, SLC-4E) within government facilities.
- **GitHub Link** -- https://nbviewer.org/github/ADVAIT135/IBM-DATA-SCIENCE/blob/ced8d356aaf8eeddd78ac496b21035df8e507601/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission/lab-jupyter-launch-site-location_final.ipynb

Build a Dashboard with Plotly Dash

- Plots and Interactions Added
- **Pie Chart: Total Successful Launches by Site**
 - Shows success distribution across KSC LC-39A, CCAFS LC-40, VAFB SLC-4E, and CCAFS SLC-40.
 - Dropdown filter allows selection of individual sites.
- **Pie Chart: Success vs Failure by Selected Site**
 - Displays launch outcome breakdown for the selected site (e.g., KSC LC-39A).
 - Interactive dropdown updates chart dynamically.
- **Scatter Plot: Payload Mass vs Launch Outcome**
- X-axis: Payload Mass (kg)
- Y-axis: Launch Outcome (Class: 0 = Failure, 1 = Success)
- Color-coded by Booster Version (v1.0, v1.1, FT, B4, B5)
- Interactive payload range slider filters data.
- Why These Plots and Interactions Were Added
- **Site-Level Analysis:**
 - Pie charts help compare reliability across launch sites and highlight operational strengths (e.g., KSC LC-39A's 76.9% success rate).
- **Booster Performance Insight:**
 - Scatter plot reveals correlation between booster evolution and success, especially for mid-to-heavy payloads.
- **User Interactivity:**
- Dropdowns and sliders allow dynamic filtering, making the dashboard exploratory and decision-supportive.
- **Git Hub Link** -- https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission/spacex-dash-app_final.py

Predictive Analysis (Classification)

➤ Data Preparation

- Target: Class column (binary: 0 = failure, 1 = success)
- Features: Scaled using StandardScaler
- Split: 80/20 train-test using train_test_split

➤ Models Built & Tuned

➤ Logistic Regression

- GridSearchCV with C, penalty, solver
- Best Accuracy: 0.8196 (CV), 0.8333 (Test)

➤ Support Vector Machine (SVM)

- RandomizedSearchCV with kernel, C, gamma
- Best Accuracy: 0.85 (CV), 0.7222 (Test)

➤ Decision Tree

- GridSearchCV with criterion, splitter, depth, etc.
- Best Accuracy: 0.8875 (CV), 0.8889 (Test)

➤ K-Nearest Neighbors (KNN)

- GridSearchCV with n_neighbors, algorithm, p
- Best Accuracy: 0.6643 (CV), 0.6111 (Test)

➤ Evaluation Metrics

- Accuracy scores computed using accuracy_score
- Confusion matrices plotted with seaborn heatmaps
- Cross-validation used to compare model stability

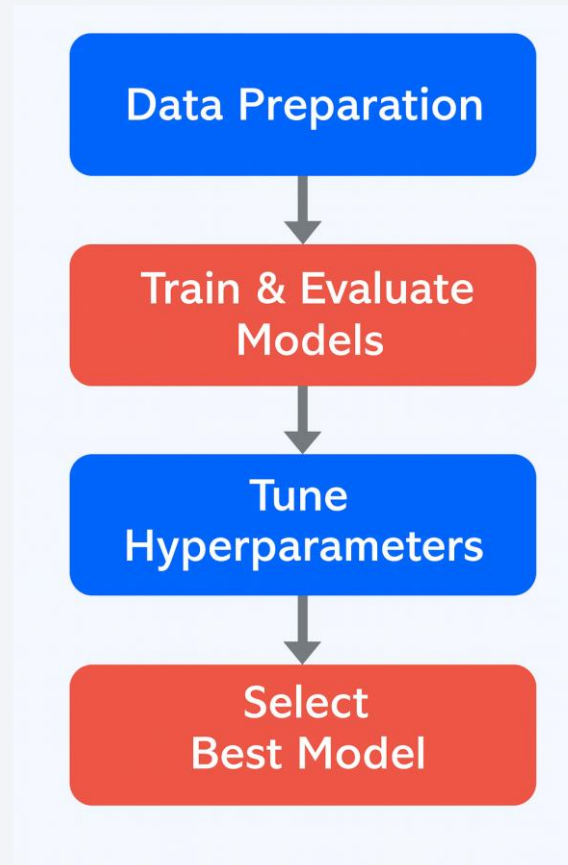
➤ Best Performing Model

- **Logistic Regression:** Highest mean accuracy (0.8196) and strong generalization

- Visualized with bar chart comparing all models

- **Git Hub Link** -- [https://github.com/ADVAIT135/IBM-DATA-](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission/SpaceX-Machine-Learning-Prediction-Final.ipynb)

[SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission/SpaceX-Machine-Learning-Prediction-Final.ipynb](https://github.com/ADVAIT135/IBM-DATA-SCIENCE/blob/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission/SpaceX-Machine-Learning-Prediction-Final.ipynb)



Results

Exploratory Data Analysis (EDA)

•Launch Trends:

- KSC LC-39A had the highest success rate.
- Early flights showed more failures; later missions improved.

•Orbit Insights:

- ES-L1, GEO, and SSO had near-perfect success.
- GTO and SO showed higher failure rates.

•Payload Patterns:

- Heavier payloads correlated with newer boosters and higher success.
- LEO and ISS had mixed outcomes across a wide payload range.

•Yearly Success Trend:

- Success rate grew steadily from 2014 to peak in 2019 (~90%).

Interactive Analytics Demo (Dash)

•Dashboard Features:

- Pie charts for site-wise success and failure breakdowns.
- Scatter plot showing payload vs. success, colored by booster version.
- Interactive dropdowns and sliders for filtering.

•Key Findings:

- KSC LC-39A had 76.9% success rate.
- FT and B5 boosters dominated successful launches in 2,000–6,000 kg range.

Mapping Insights (Folium)

•Launch Site Visualization:

- Mapped major facilities in California (VAFB) and Florida (CCAFS, KSC).
- Cluster circles showed facility density: 46 in Florida vs. 10 in California.

•Strategic Geography:

- Florida supports equatorial and crewed missions; California specializes in polar orbits.
- Proximity to airports, highways, and railways enhances logistics and launch readiness.

Predictive Analysis (Classification)

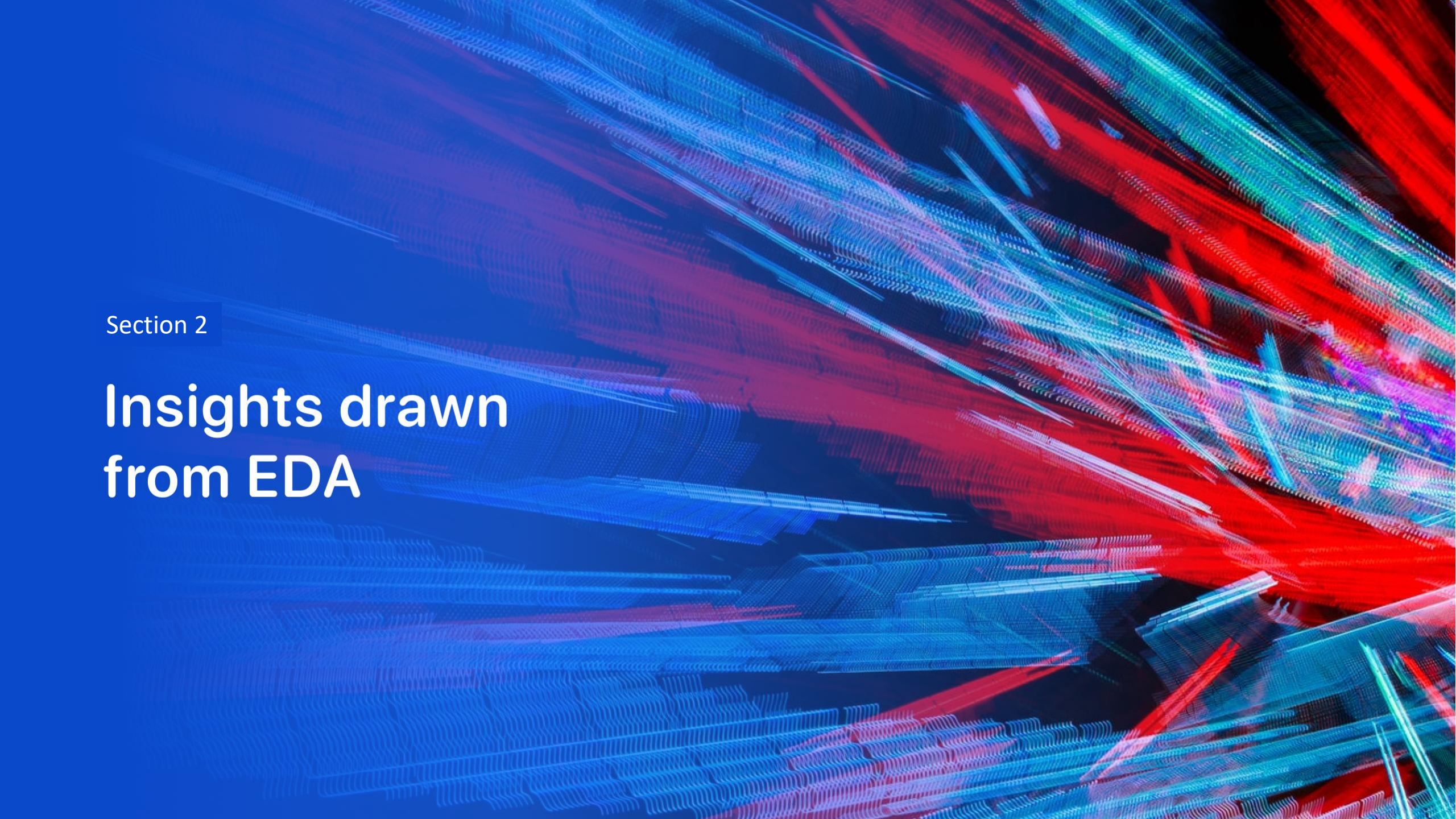
•Models Compared: Logistic Regression, SVM, Decision Tree, KNN

•Best Model:

- **Logistic Regression** with 0.8196 mean accuracy and 0.8333 test accuracy.
- Confusion matrix showed perfect recall for successful landings.

•Use Case:

- Ideal for minimizing missed successful landings, even with tolerable false positives.



Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Axes:**

- X-axis:** Flight Number (chronological order of launches)

- Y-axis:** Launch Site (CCAFS SLC 40, VAFB SLC 4E, KSC LC 39A)

- Color Coding (Hue = Class):**

- Orange (Class 1):** Successful launches

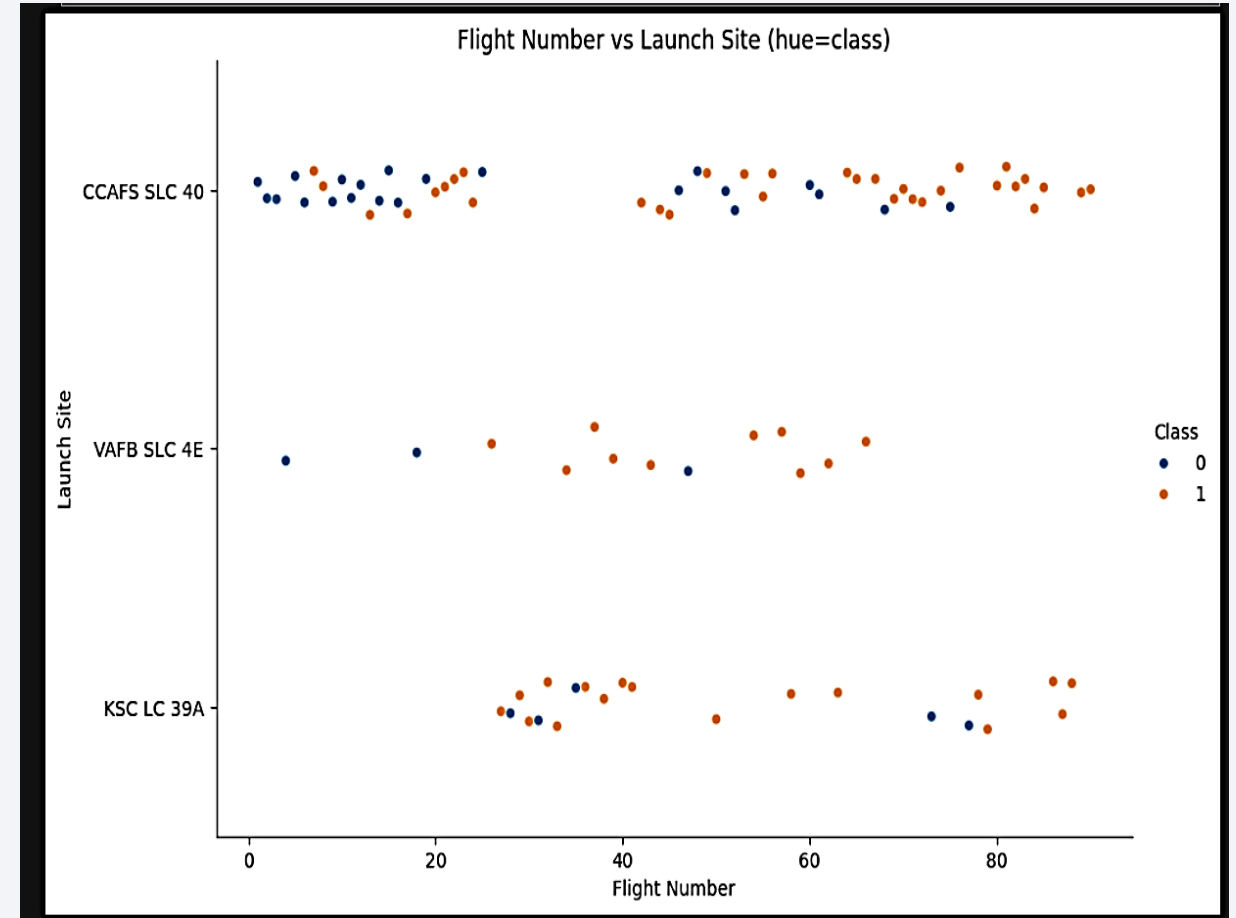
- Blue (Class 0):** Failed launches

- Insights:**

- Launch outcomes vary across sites and over time.

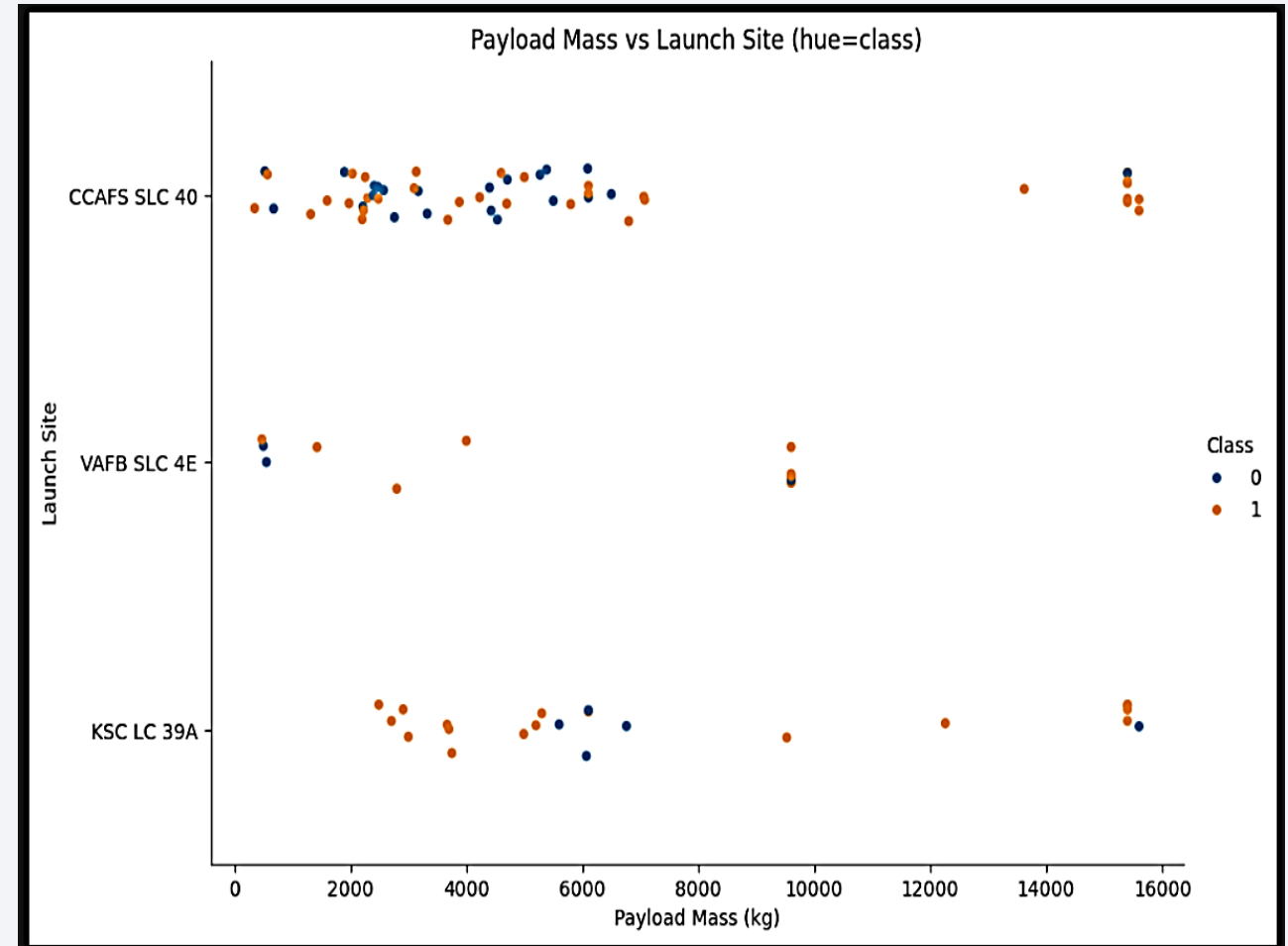
- KSC LC 39A shows a higher concentration of successful launches.

- Early flight numbers have more failures, suggesting improvement over time.



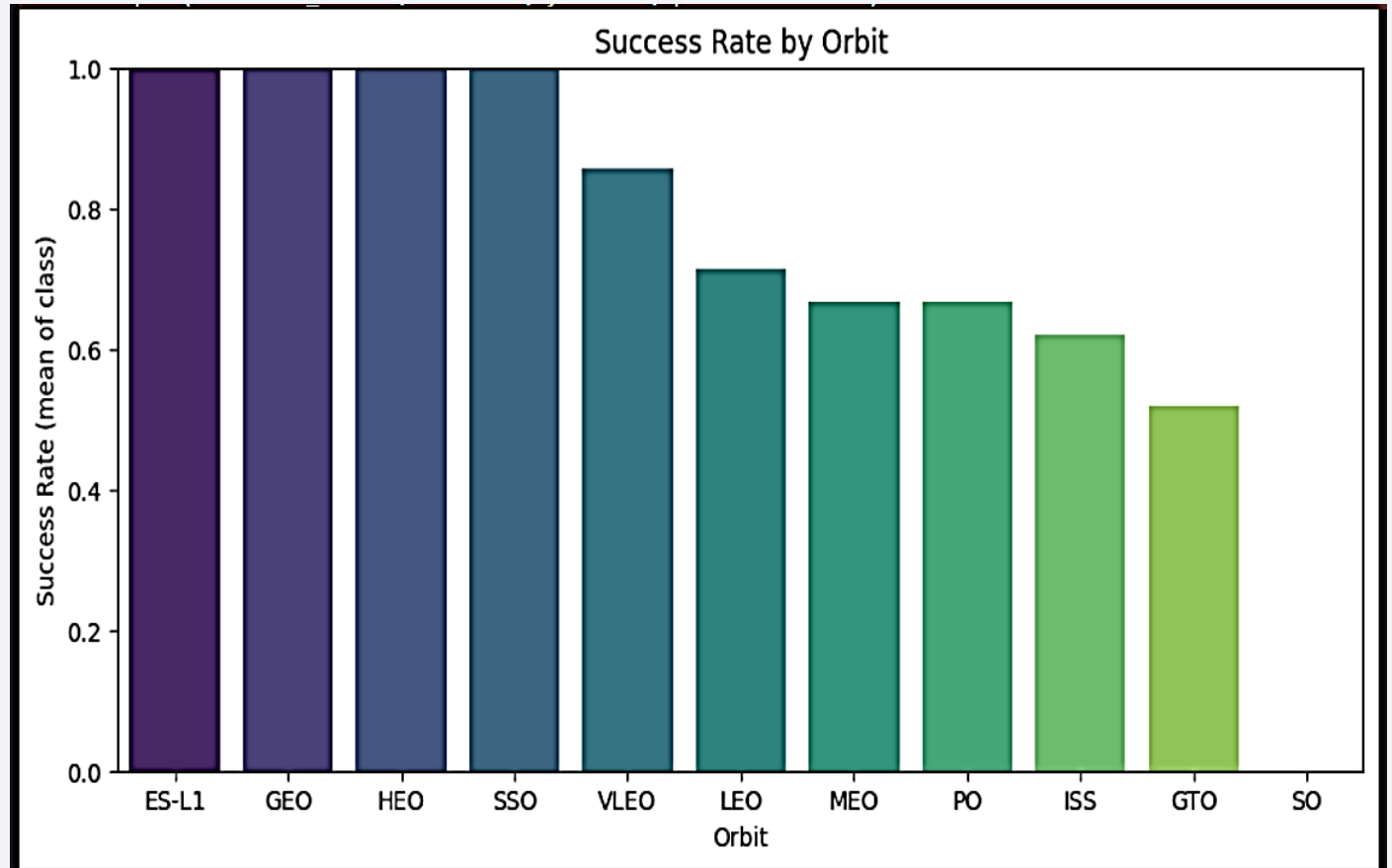
Payload vs. Launch Site

- **X-axis:** Payload Mass (kg)
- **Y-axis:** Launch Site (CCAFS SLC 40, VAFB SLC 4E, KSC LC 39A)
- **Color Coding (Hue = Class):**
 - **Orange (Class 1):** Successful launches
 - **Blue (Class 0):** Failed launches
- **Insights:**
 - KSC LC 39A handled heavier payloads with mostly successful outcomes.
 - VAFB SLC 4E shows fewer launches and more failures.
 - CCAFS SLC 40 has a mix of outcomes across a wide payload range.



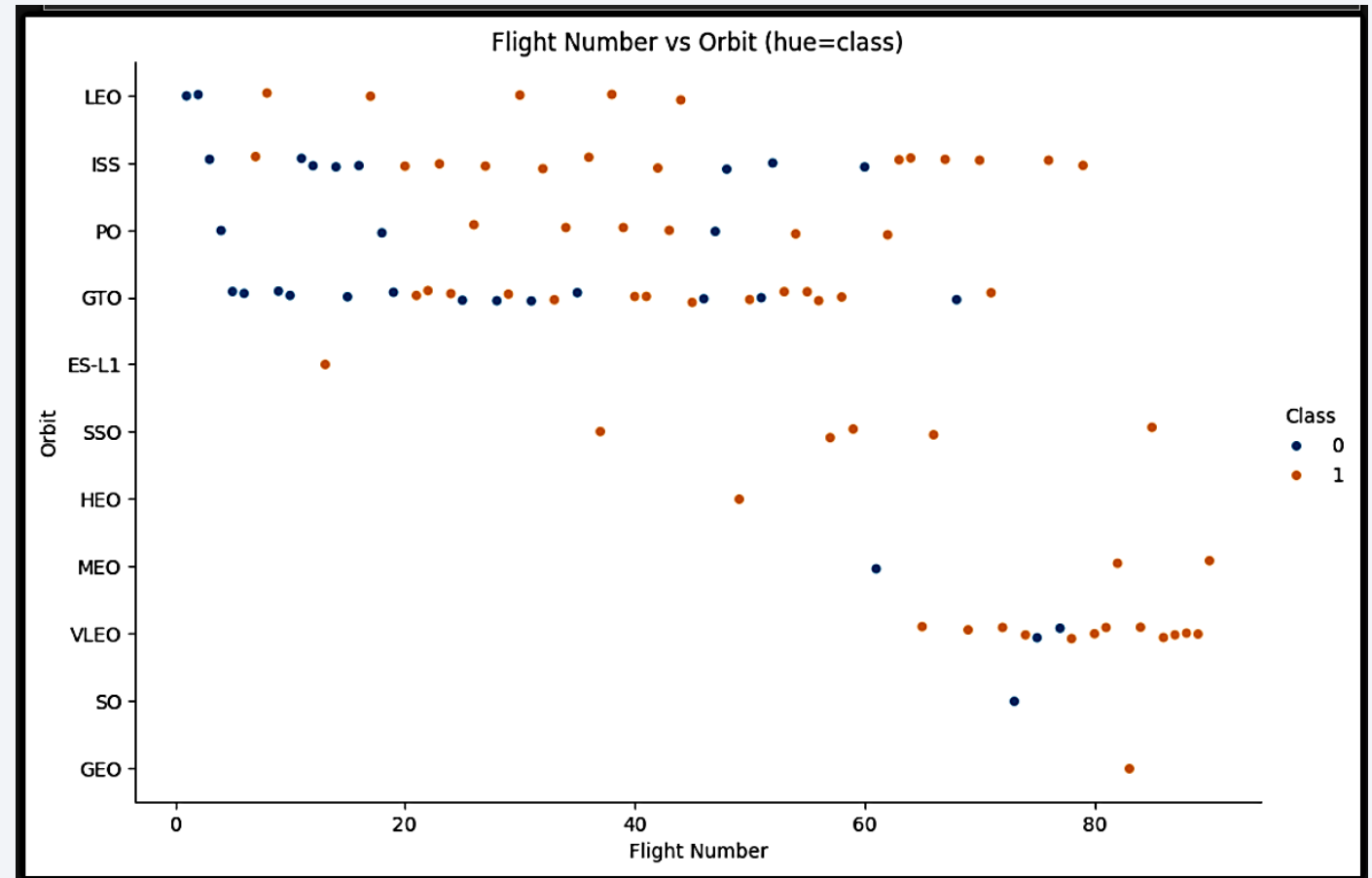
Success Rate vs. Orbit Type

- X-axis:** Orbit Types (e.g., ES-L1, GEO, LEO, GTO)
- Y-axis:** Success Rate (Mean of Class, ranging from 0.0 to 1.0)
- Color Bars Represent:** Average success rate of launches to each orbit
 - Class 1 = Success, Class 0 = Failure**
- Key Insights:**
 - Highest Success:** ES-L1, GEO, HEO, and SSO — nearly perfect launch records
 - Moderate Success:** LEO, MEO, PO — mixed outcomes
 - Lowest Success:** SO and GTO — higher failure rates, indicating greater mission risk



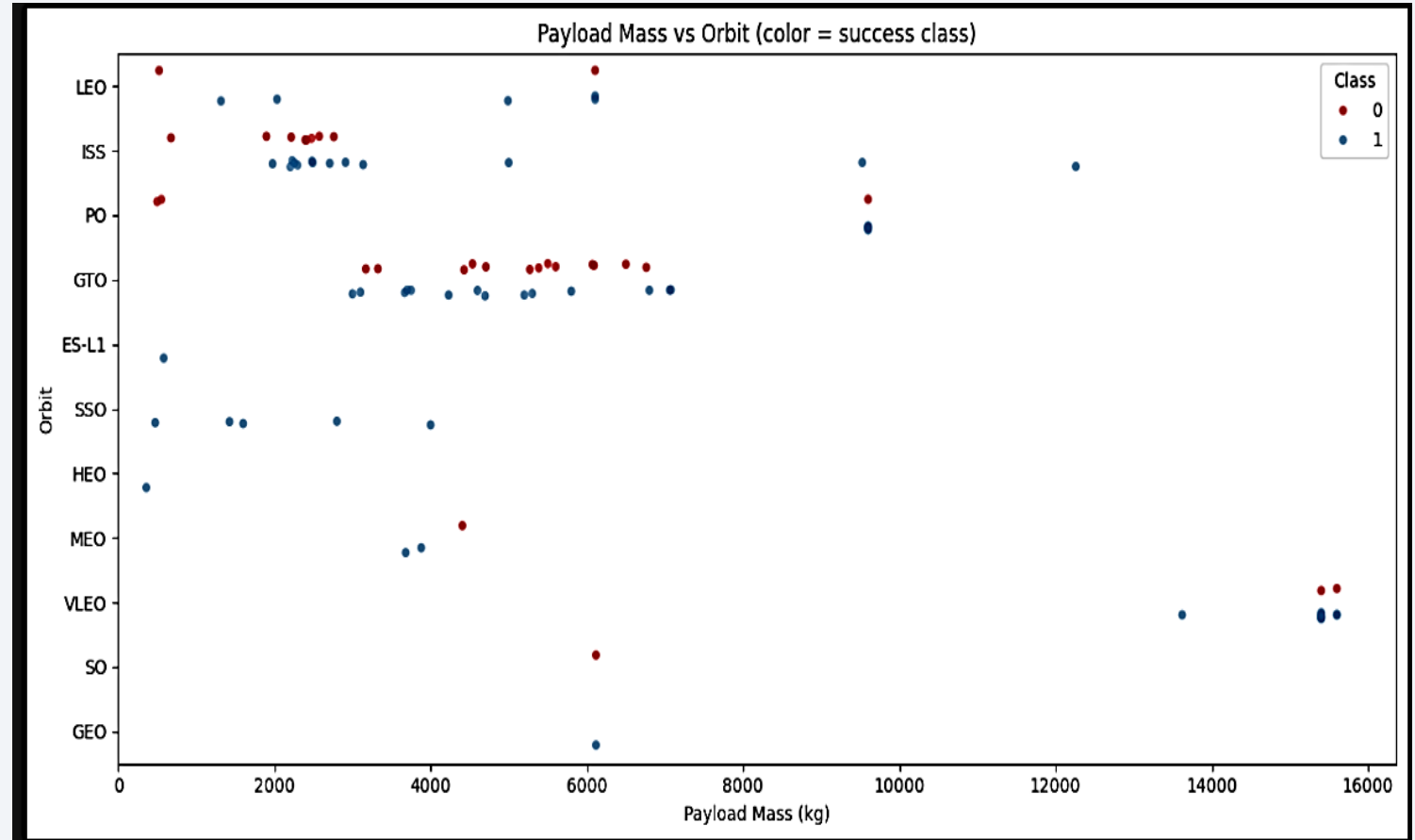
Flight Number vs. Orbit Type

- **X-axis:** Flight Number (chronological order of launches)
- **Y-axis:** Orbit Type (e.g., LEO, GTO, ISS, GEO, ES-L1)
- **Color Coding (Hue = Class):**
 - **Orange (Class 1):** Successful launches
 - **Blue (Class 0):** Failed launches
- **Key Insights:**
 - **LEO and ISS** have frequent launches with mixed outcomes
 - **GTO and SO** show higher failure rates
 - **High success** observed for ES-L1, GEO, and SSO missions
 - Later flight numbers trend toward more successful outcome



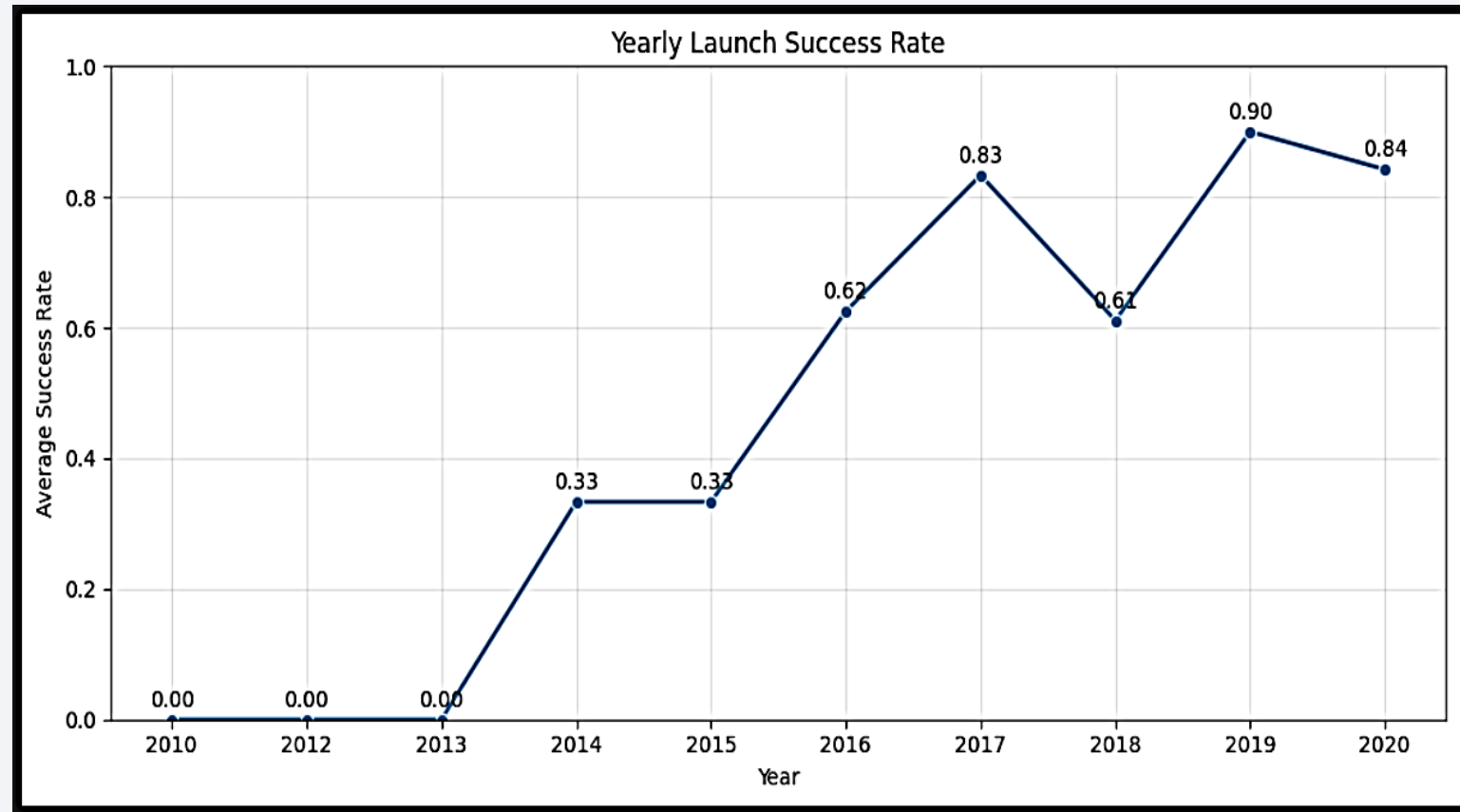
Payload vs. Orbit Type

- **X-axis:** Payload Mass (kg)
- **Y-axis:** Orbit Type (LEO, GTO, ISS, GEO, etc.)
- **Color Coding (Success Class):**
 - **Blue (Class 1):** Successful launches
 - **Red (Class 0):** Failed launches
- **Key Insights:**
 - **Higher payloads** tend to be associated with successful missions to ES-L1, GEO, and SSO
 - **GTO and SO** orbits show more failures, especially with heavier payloads
 - **LEO and ISS** have a wide payload range with mixed outcomes
 - Suggests orbit type and payload mass jointly influence launch success



Launch Success Yearly Trend

- X-axis:** Year (2010 to 2020)
- Y-axis:** Average Success Rate (0.0 to 1.0)
- Trend Overview:**
 - Early years (2010–2013): **0% success rate**
 - Gradual improvement from 2014 onward
 - Peak success** in 2019 at **90%**, followed by a slight dip in 2020
- Key Insight:**
 - The plot reflects **consistent growth in launch reliability**, indicating technological and operational advancements over the decade



All Launch Site Names

```
In [11]: %sql select distinct "Launch_Site" from spacetable
```

- Purpose:**

Retrieve a list of **unique launch sites** from the spacetable dataset.

- Key Components:**

- SELECT DISTINCT: Ensures **no duplicates** in the result
- "Launch_Site": The column being queried
- FROM spacetable: Specifies the source table

- Use Case:**

Useful for identifying all launch locations used in the dataset — ideal for filtering, grouping, or visualizing launch site activity.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Purpose:**

Retrieve the **first 5 records** from spacetable where the launch site name begins with **"CCA"**.

- Key Components:**

- SELECT *:** Returns **all columns** of matching rows
- WHERE "Launch_Site" LIKE 'CCA%':** Filters rows where launch site starts with "CCA" (e.g., CCAFS SLC 40)
- LIMIT 5:** Restricts output to **5 rows only**

- Use Case:**

Useful for **previewing data** from a specific launch site pattern without overwhelming results.

```
In [12]: %sql select * from spacetable where "Launch_Site" like "CCA%" limit 5
```

```
* sqlite:///my_data1.db
```

Done.										
Out[12]:	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
In [13]: %sql select sum("PAYLOAD_MASS_KG_") from spacetable where Customer is "NASA (CRS)"
* sqlite:///my_data1.db
Done.

Out[13]: sum("PAYLOAD_MASS_KG_")
          45596
```

- Purpose:**

Calculate the **total payload mass** (in kilograms) launched by **NASA (CRS)** missions.

- Key Components:**

- SUM("PAYLOAD_MASS_KG_"): Adds up all payload masses
- FROM spacetable: Uses the SpaceX launch dataset
- WHERE Customer IS 'NASA (CRS)': Filters for NASA (CRS) launches only

- Result:**

Total Payload Mass = 45,596 kg

- Use Case:**

Useful for analyzing NASA's cargo contribution and mission scale over time.

Average Payload Mass by F9 v1.1

```
In [14]: %sql select avg("PAYLOAD_MASS_KG_") from spacetable where "Booster_Version" is "F9 v1.1"
* sqlite:///my_data1.db
Done.
Out[14]: avg("PAYLOAD_MASS_KG_")
          2928.4
```

- Purpose:**

Calculate the **average payload mass** (in kilograms) for launches using the **F9 v1.1** booster version.

- Key Components:**

- AVG("PAYLOAD_MASS_KG_"): Computes the mean payload mass
- FROM spacetable: Uses the SpaceX launch dataset
- WHERE "Booster_Version" IS 'F9 v1.1': Filters for a specific booster type

- Result:**

Average Payload Mass = 2,928.4 kg

- Use Case:**

Useful for comparing booster performance and payload capacity across different rocket versions.

First Successful Ground Landing Date

```
In [16]: %sql select min(Date) from spacetable where "Landing_Outcome" is "Success (ground pad)"
* sqlite:///my_data1.db
Done.
Out[16]: min(Date)
          2015-12-22
```

- Purpose:**

Identify the **earliest date** when a rocket successfully landed on a **ground pad**.

- Key Components:**

- MIN(Date): Finds the **earliest date** in the dataset
- FROM spacetable: Uses the SpaceX launch records
- WHERE "Landing_Outcome" IS 'Success (ground pad)': Filters for successful ground pad landings only

- Result:**

First Ground Pad Success = 2015-12-22

- Use Case:**

Useful for tracking the **start of successful landing milestones** in SpaceX's reusability journey.

a

Successful Drone Ship Landing with Payload between 4000 and 6000

- **SELECT DISTINCT "Booster_Version"**
FROM spacetable
WHERE "Landing_Outcome" IS 'Success (drone ship)'
AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000
- **Purpose:**
Identify **unique booster versions** that successfully landed on a **drone ship** while carrying payloads between **4000–6000 kg**.
- **Key Components:**
 - **SELECT DISTINCT:** Ensures **no duplicate booster versions**
 - **WHERE "Landing_Outcome" IS 'Success (drone ship)':** Filters for successful drone ship landings
 - **AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000:** Narrows results to medium-weight payloads
- **Use Case:**
Helps analyze **booster performance** under specific landing and payload conditions — useful for mission planning and hardware evaluation.

```
Done.  
Out[19]: Booster_Version  
          F9 FT B1022  
          F9 FT B1026  
          F9 FT B1021.2  
          F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

```
In [21]: %%sql SELECT "Landing_Outcome", COUNT(*) AS total
        FROM spacetable
        WHERE "Landing_Outcome" IN ('Success', 'Failure')
        GROUP BY "Landing_Outcome";

* sqlite:///my_data1.db
Done.
```

```
Out[21]: Landing_Outcome  total
         -----
         Failure         3
         Success        38
```

- Purpose:**

Count the number of launches with outcomes labeled '**Success**' or '**Failure**'.

- Key Components:**

- SELECT "Landing_Outcome", COUNT(*): Retrieves each outcome and its total count
- WHERE ... IN ('Success', 'Failure'): Filters only for success and failure records
- GROUP BY "Landing_Outcome": Aggregates results by each outcome type

- Result:**

✓ **Success:** 38 launches

✗ **Failure:** 3 launches

- Use Case:**

Useful for assessing **overall landing reliability** and identifying performance trends.

Boosters Carried Maximum Payload

```
In [22]: %%sql SELECT DISTINCT "Booster_Version"
FROM spacetable
WHERE "PAYLOAD_MASS__KG_" = (
    SELECT MAX("PAYLOAD_MASS__KG_") FROM spacetable
);
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[22]: Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

- Purpose:**

Identify all **booster versions** that carried the **heaviest payload** ever recorded in the dataset.

- Key Components:**

- SELECT DISTINCT: Ensures **no duplicate booster entries**
- WHERE "PAYLOAD_MASS__KG_" = (...): Filters rows matching the **maximum payload mass**
- Subquery: SELECT MAX("PAYLOAD_MASS__KG_"): Finds the **largest payload value**

- Result:**

✓ Returns a list of boosters (e.g., F9 B5 B1048.4, B1051.6, etc.) that handled the **maximum payload**

- Use Case:**

Useful for analyzing **top-performing boosters** in terms of payload capacity — valuable for mission planning and hardware benchmarking.

2015 Launch Records

```
In [23]: %%sql SELECT
CASE substr("Date", 6, 2)
  WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March'
  WHEN '04' THEN 'April'   WHEN '05' THEN 'May'       WHEN '06' THEN 'June'
  WHEN '07' THEN 'July'    WHEN '08' THEN 'August'    WHEN '09' THEN 'September'
  WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December'
END AS month_name,
"Landing_Outcome",
"Booster_Version",
"Launch_Site"
FROM spacetable
WHERE substr("Date", 1, 4) = '2015'
AND "Landing_Outcome" = 'Failure (drone ship)';

* sqlite:///my_data1.db
Done.
```

```
Out[23]:
```

month_name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Purpose:**

Extract details of **failed drone ship landings** in **2015**, with month names derived from the date.

- Key Components:**

- CASE substr("Date", 6, 2): Converts numeric month to **month name**
- WHERE substr("Date", 1, 4) = '2015': Filters for **year 2015**
- AND "Landing_Outcome" = 'Failure (drone ship)': Focuses on **specific failure type**

- Result:**

Displays **month**, **landing outcome**, **booster version**, and **launch site** for each failed attempt.

- Use Case:**

Useful for identifying **seasonal patterns** or operational issues in failed drone ship landings.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
In [24]: %%sql SELECT "Landing_Outcome",
          COUNT(*) AS outcome_count
          FROM spacetable
          WHERE substr("Date", 1, 10) BETWEEN '2010-06-04' AND '2017-03-20'
          GROUP BY "Landing_Outcome"
          ORDER BY outcome_count DESC;

* sqlite:///my_data1.db
Done.

Out[24]:
```

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Purpose:**

Count and rank different **landing outcomes** for launches between **June 2010 and March 2017**.

- Key Components:**

- substr("Date", 1, 10): Extracts the full date string
- BETWEEN '2010-06-04' AND '2017-03-20': Filters for a specific time range
- GROUP BY "Landing_Outcome": Aggregates counts by outcome type
- ORDER BY outcome_count DESC: Sorts results from **most to least frequent**

- Result Highlights:**

- Most common:** No attempt (10 launches)
- Successful drone ship landings:** 5
- Failures:** Drone ship (5), parachute (2), and others

- Use Case:**

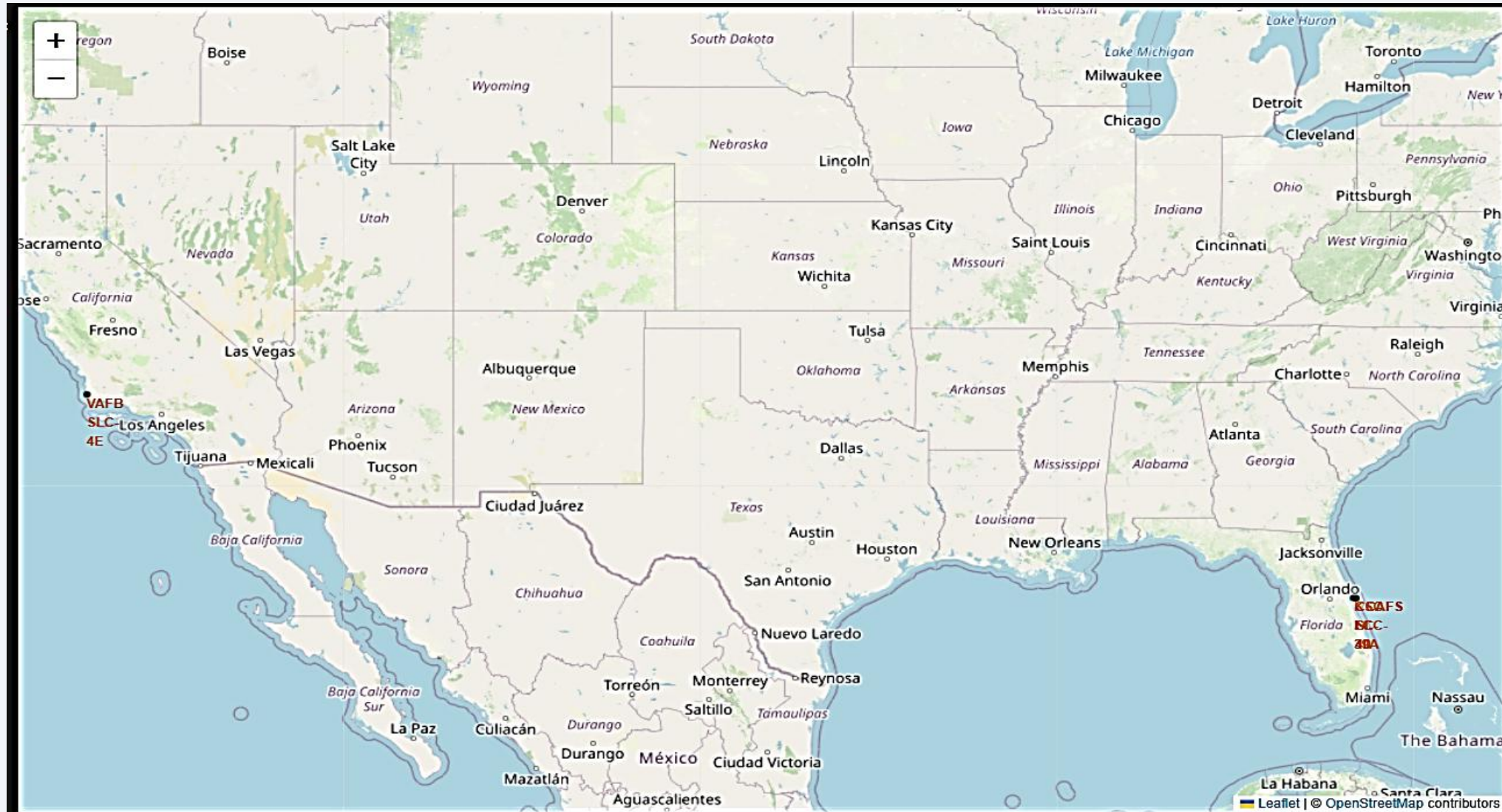
Useful for evaluating **landing strategy trends** and identifying areas for improvement during early SpaceX missions.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

Major U.S. Space Launch Facilities: California and Florida



Major U.S. Space Launch Facilities: California and Florida

- Key Elements of the Map
- This map highlights **strategic space launch sites** in the southern United States, specifically in **California** and **Florida**, which are the two primary hubs for American space missions.
- California Launch Sites
- **VAFB (Vandenberg Air Force Base)**: Located on the central coast, VAFB is crucial for launching satellites into polar or sun-synchronous orbits. It supports both military and commercial launches.
- **SLC (Space Launch Complex)** and **4E**: These are specific pads within VAFB used for different launch vehicles, including Falcon 9 and other rockets.
- Florida Launch Sites
- **KSC/AFS (Kennedy Space Center / Cape Canaveral Air Force Station)**: This is the heart of NASA's crewed spaceflight operations, including Apollo and Artemis missions.
- **ECC (Eastern Control Center)**: Manages launch operations and telemetry for missions from the Eastern Range.
- **39A**: One of the most iconic launch pads, used for Apollo missions and now leased by SpaceX for Falcon 9 and Falcon Heavy launches.

Major U.S. Space Launch Facilities: California and Florida

- Findings and Insights
- **Geographic Advantage:** Florida's east coast enables launches into equatorial orbits, while California supports polar trajectories. This dual setup allows the U.S. to cover a wide range of orbital needs.
- **Public-Private Collaboration:** Sites like 39A and SLC 4E are now operated by private companies (e.g., SpaceX), reflecting the shift toward commercial spaceflight.
- **National Security and Science:** These facilities support both civilian (NASA) and defense (U.S. Space Force) missions, underlining their strategic importance.

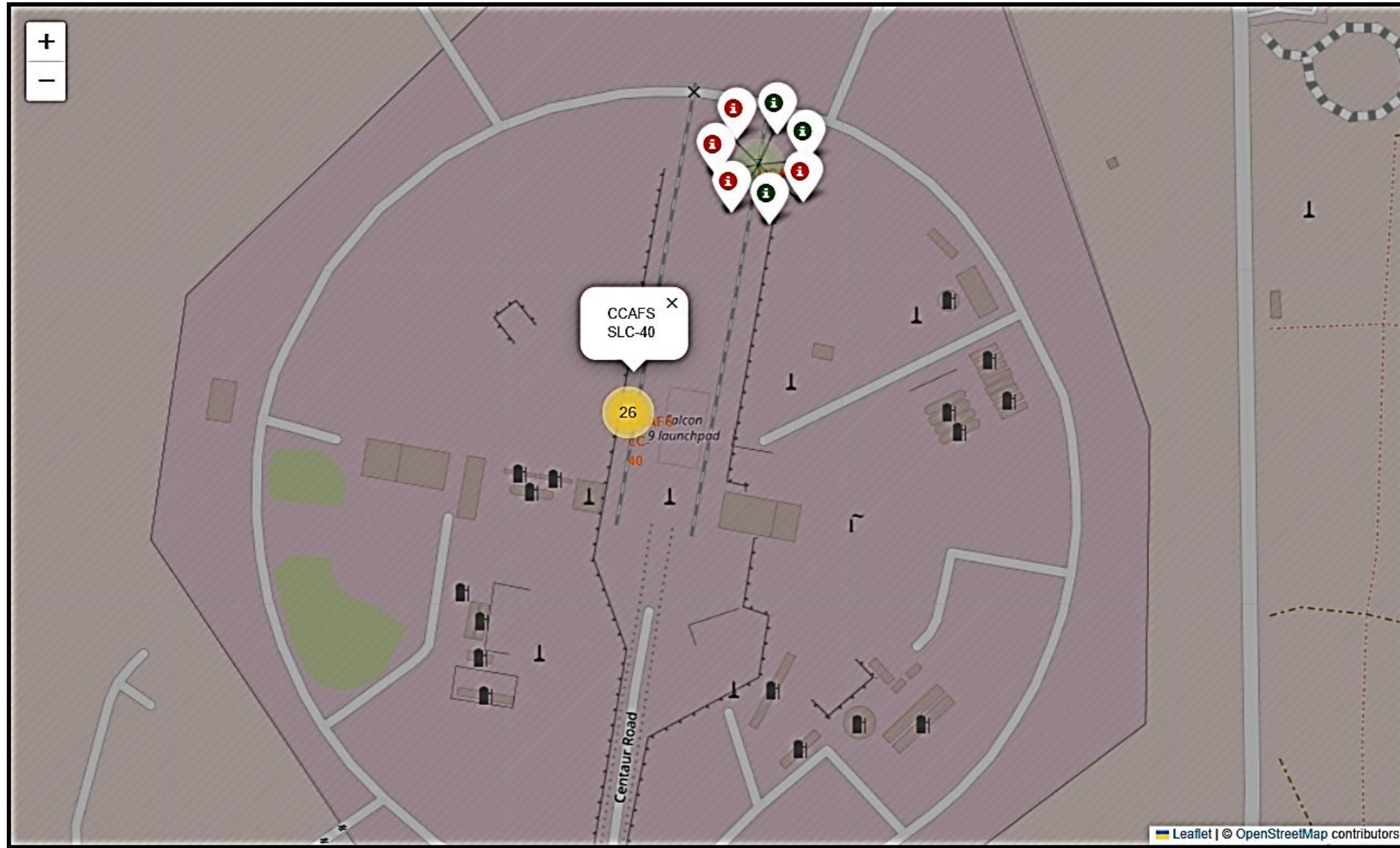
Strategic U.S. Launch Operations: VAFB and SFS Facility Clusters

- Key Elements of the Map
- This map showcases **two major clusters of aerospace and launch infrastructure** in the United States—one in **California** and one in **Florida**—each marked with a yellow circle and a numeric label indicating the number of associated facilities.
- California Cluster (Label: 10)
- **VAFB (Vandenberg Air Force Base)**: A key site for polar orbit launches, used by both military and commercial entities.
- **SLC (Space Launch Complex)**: Refers to specific launch pads within VAFB, such as SLC-4E, used by SpaceX and others.
- **4E**: Likely denotes SLC-4E, a pad used for Falcon 9 launches into polar orbits.
- Florida Cluster (Label: 46)
- **SFS (Space Force Station)**: Formerly Cape Canaveral Air Force Station, now operated by the U.S. Space Force.
- **ECC (Eastern Control Center)**: Manages telemetry and command operations for launches from the Eastern Range.
- **20A**: Refers to Launch Complex 20A, part of the broader Cape Canaveral infrastructure.

Strategic U.S. Launch Operations: VAFB and SFS Facility Clusters

- Findings and Insights
- **Facility Density:** Florida hosts a significantly larger number of launch-related facilities (46) compared to California (10), reflecting its dominance in equatorial and crewed mission launches.
- **Operational Specialization:**
 - **California** specializes in polar and sun-synchronous orbits, ideal for Earth observation and reconnaissance.
 - **Florida** supports a broader range of missions, including crewed flights, deep space missions, and commercial launches.
- **Military and Civilian Integration:** Both clusters include facilities operated by the U.S. Space Force and NASA, highlighting the integration of defense and civilian space programs.
- **Geographic Strategy:** The coastal locations minimize risk to populated areas and optimize launch trajectories.

Cape Canaveral Launch Infrastructure: Focus on SLC-40 and Falcon 9 Operations



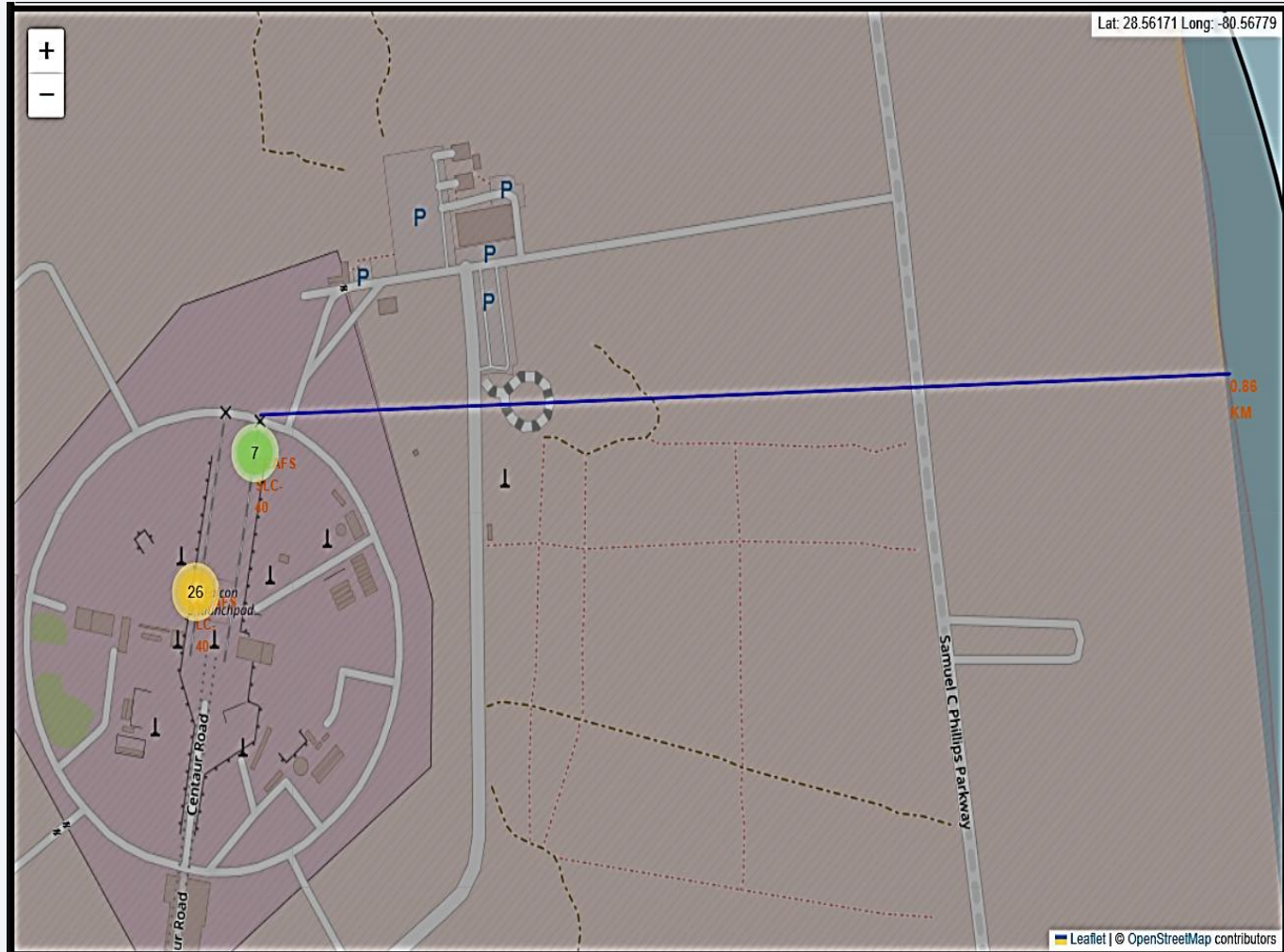
Cape Canaveral Launch Infrastructure: Focus on SLC-40 and Falcon 9 Operations

- Key Elements of the Map
- This map provides a detailed view of **Space Launch Complex 40 (SLC-40)** within **Cape Canaveral Space Force Station (CCSFS)**, a critical site for U.S. space launch operations.
- Primary Features
 - **CCAFS SLC-40**: Highlighted with marker "26", this is the dedicated launchpad for **SpaceX's Falcon 9** missions. It supports a wide range of payloads including satellites, cargo resupply missions to the ISS, and commercial launches.
 - **Centaur Road**: A key access route within the complex, often used for logistical movement and support vehicles.
 - **Information Markers**: Clustered icons near the pad indicate auxiliary facilities such as fuel storage, telemetry stations, and ground support equipment.
- Map Design Elements
 - **Shaded Zones**: Represent restricted or operational areas, emphasizing the secure and specialized nature of launch infrastructure.
 - **Leaflet & OpenStreetMap Integration**: Indicates the map is built on open-source geospatial tools, useful for public visualization and planning.

Cape Canaveral Launch Infrastructure: Focus on SLC-40 and Falcon 9 Operations

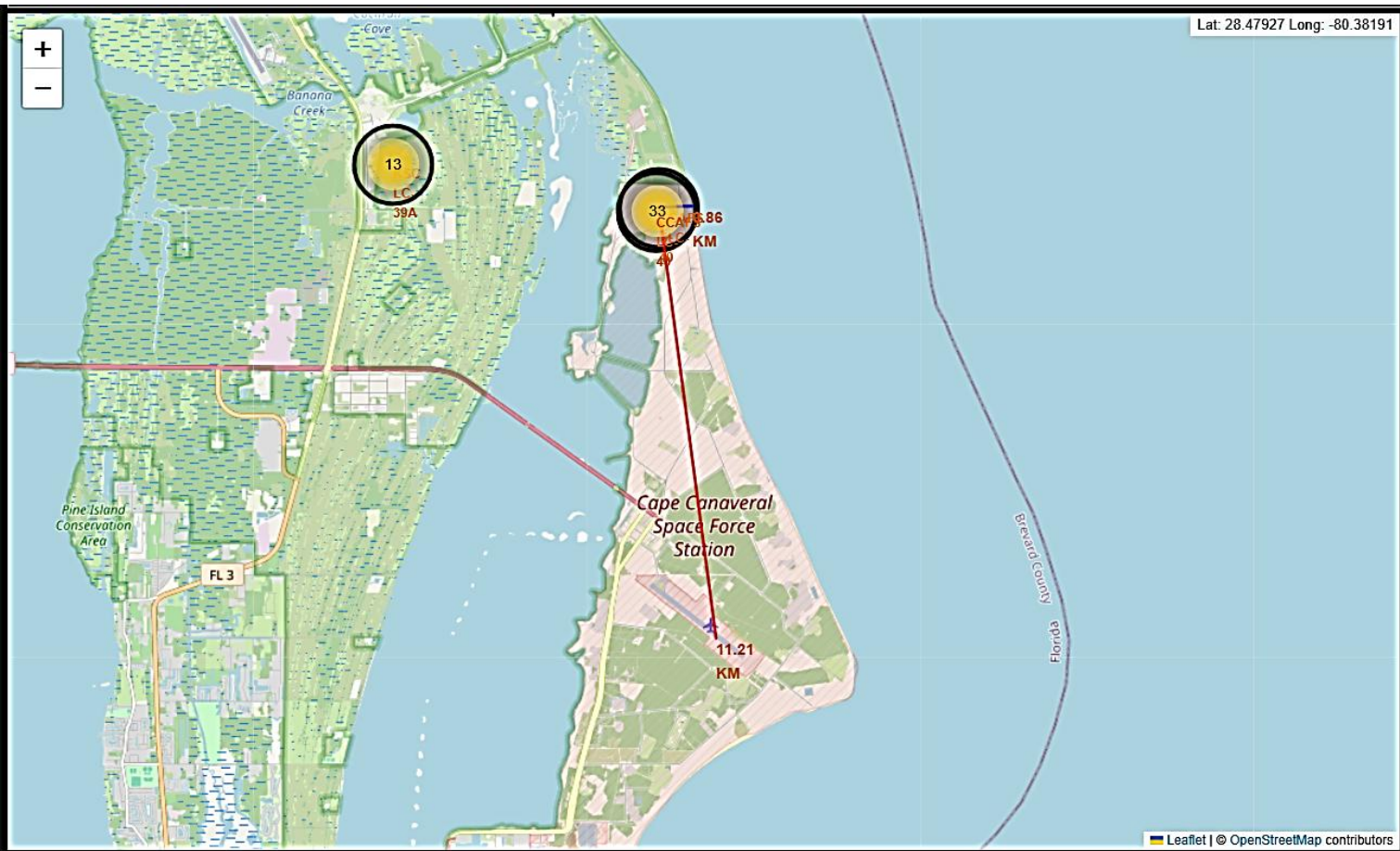
- Findings and Strategic Insights
- **Launchpad Specialization:** SLC-40 is optimized for Falcon 9, making it a high-frequency launch site for SpaceX's commercial and government missions.
- **Infrastructure Density:** The presence of multiple support structures around the pad suggests robust ground systems for fueling, telemetry, and safety.
- **Operational Efficiency:** The layout supports rapid turnaround between launches, a key factor in SpaceX's reusability strategy.
- **Geographic Advantage:** Located on Florida's east coast, the site benefits from proximity to the equator, allowing efficient launches into geostationary and low Earth orbits.

CCAFS SLC-40 Launch Site Map Overview



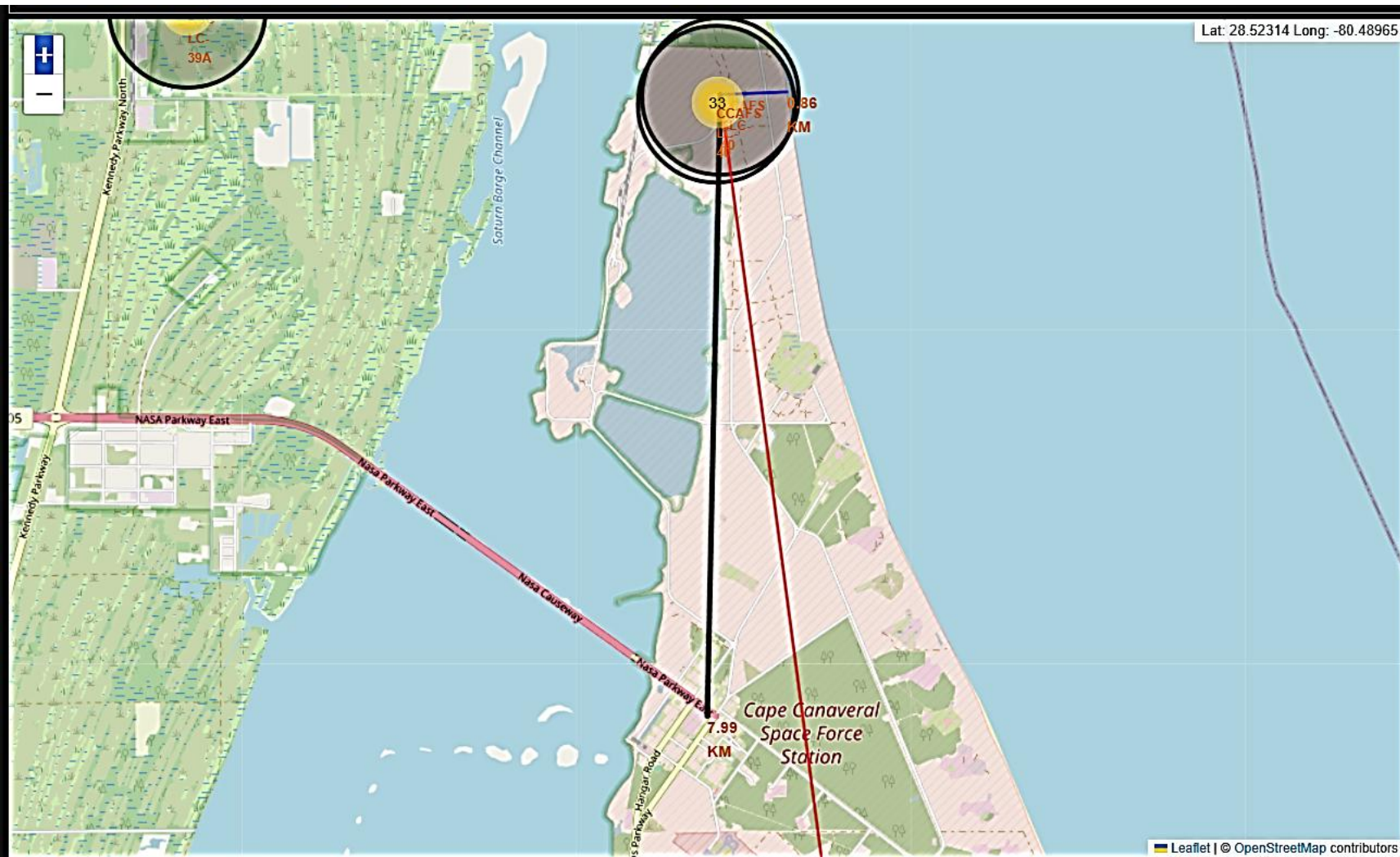
- SLC-40 & SpaceX Launchpad:** Central launch site used for Falcon 9 missions.
- AFS (Air Force Station):** Indicates military oversight, part of Cape Canaveral.
- Centaur Road & Samuel C Phillips Parkway:** Main access routes within the facility.
- Markers (e.g., 7, 26):** Likely denote observation points or operational zones.
- Parking Zones ("P"):** Designated areas for staff and visitors.
- Coordinates:** Lat 28.563, Long -80.5672 — confirms Florida location.
- Distance Line (0.86 KM):** Measures proximity from launch pad to coastline — relevant for safety and trajectory planning.

SLC-40 to Closest Airport: Spatial Snapshot



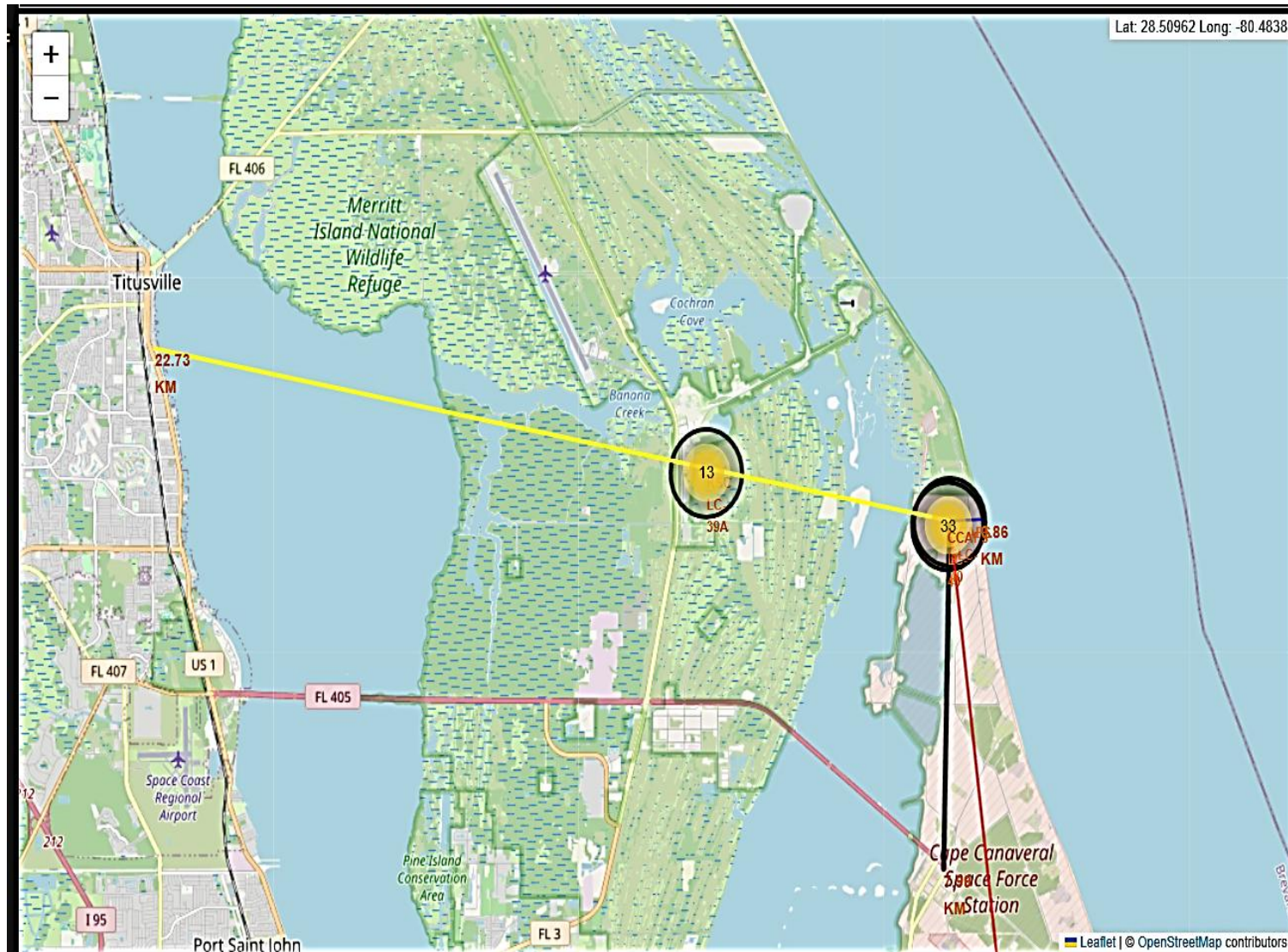
- **Launch Site:** Space Launch Complex 40 (Cape Canaveral Space Force Station)
- **Closest Airport:** Space Coast Regional Airport (KTIX), Titusville, FL
- **Straight-Line Distance:** ~11.2 km (6.96 miles)
- **Orientation:** Airport lies **west-southwest** of SLC-40
- **Road Distance:** ~15–18 km via FL-405 or FL-3
- **Strategic Relevance:**
 - Supports crew transport, logistics, and emergency ops
 - Key asset for launch readiness and recovery

SLC-40 to Closest Highway: Spatial Snapshot



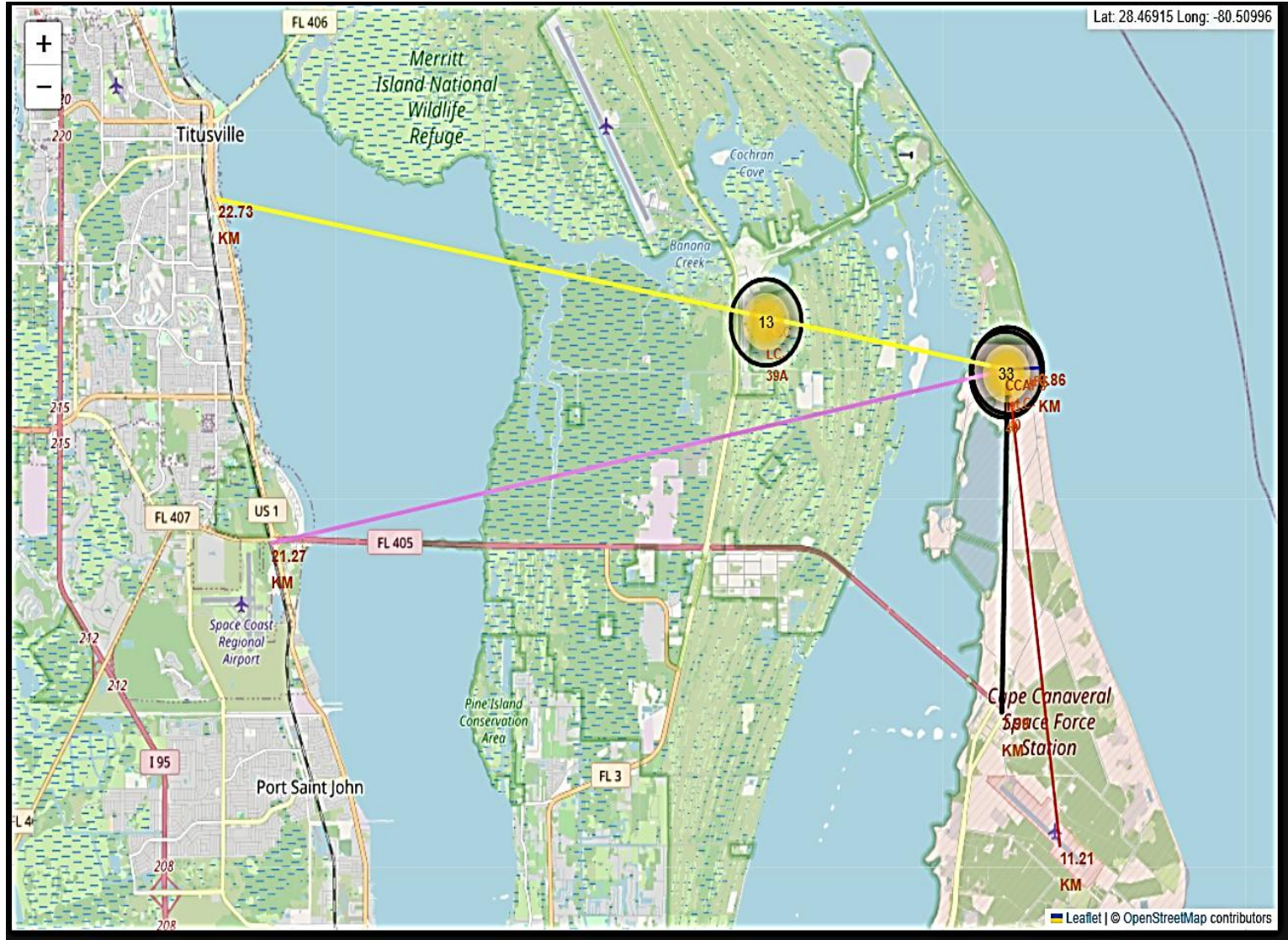
- **Launch Site:** Space Launch Complex 40 (Cape Canaveral Space Force Station)
- **Closest Major Highway:** NASA Parkway East (FL-405)
- **Straight-Line Distance:** ~8.0 km (4.97 miles)
- **Orientation:** FL-405 lies **west-northwest** of SLC-40, across the Indian River
- **Road Access:** Via Kennedy Parkway North → NASA Causeway → FL-405
- **Strategic Relevance:**
 - Primary route for **logistics, crew transport, and emergency access**
 - Connects CCAFS to **Titusville and I-95 corridor**

SLC-40 to Closest City: Spatial Snapshot

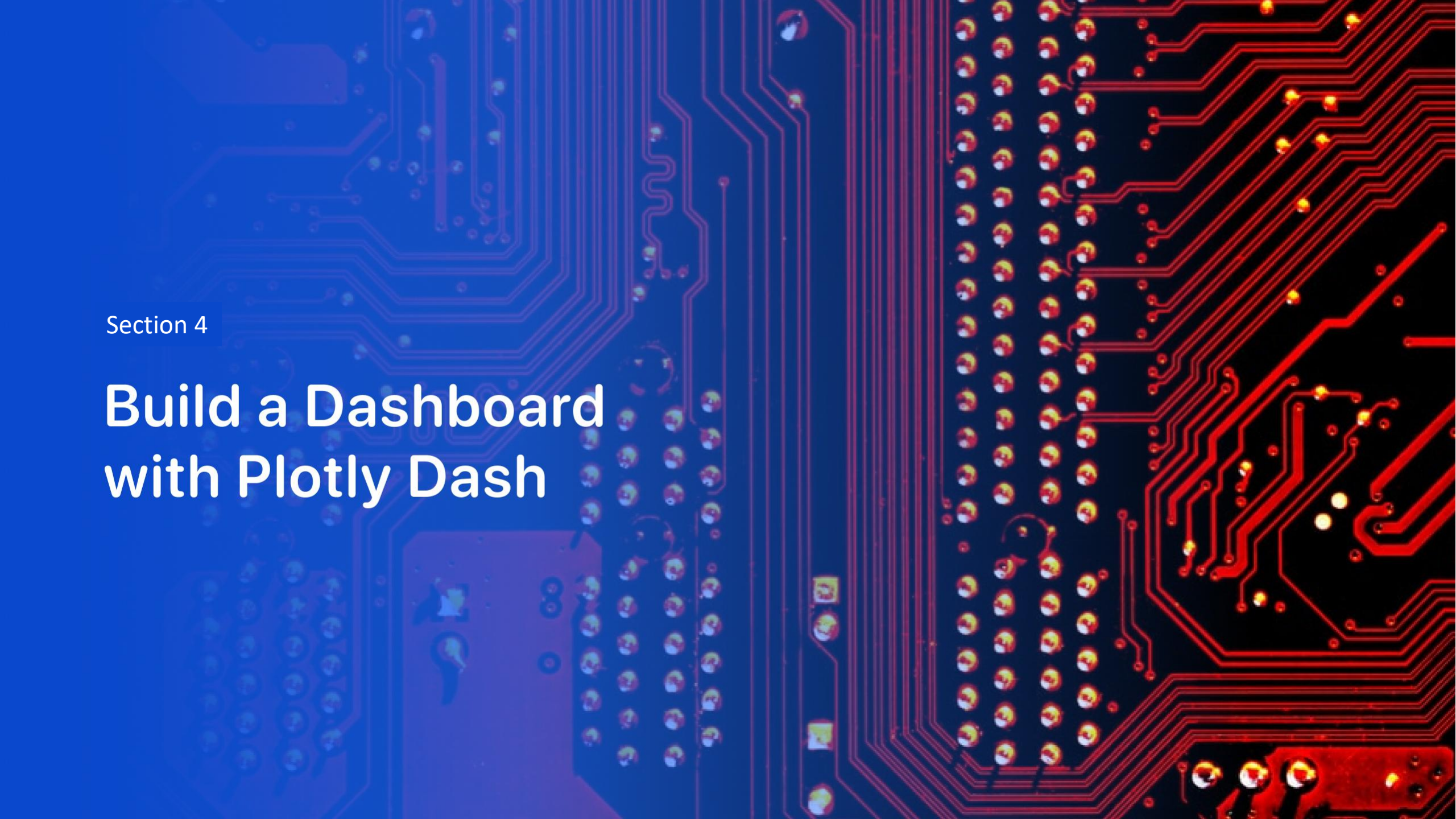


- **Launch Site:** Space Launch Complex 40 (Cape Canaveral Space Force Station)
- **Closest City:** Titusville, Florida
- **Straight-Line Distance:** ~22.7 km (14.1 miles)
- **Orientation:** Titusville lies northwest across the Indian River
- **Road Distance:** ~25–30 km via NASA Parkway East (FL-405)
- **Strategic Relevance:**
 - Titusville serves as a **logistics and lodging hub** for launch operations
 - Hosts **Space View Park**, a popular public launch viewing site

SLC-40 to Closest Railways: Spatial Snapshot



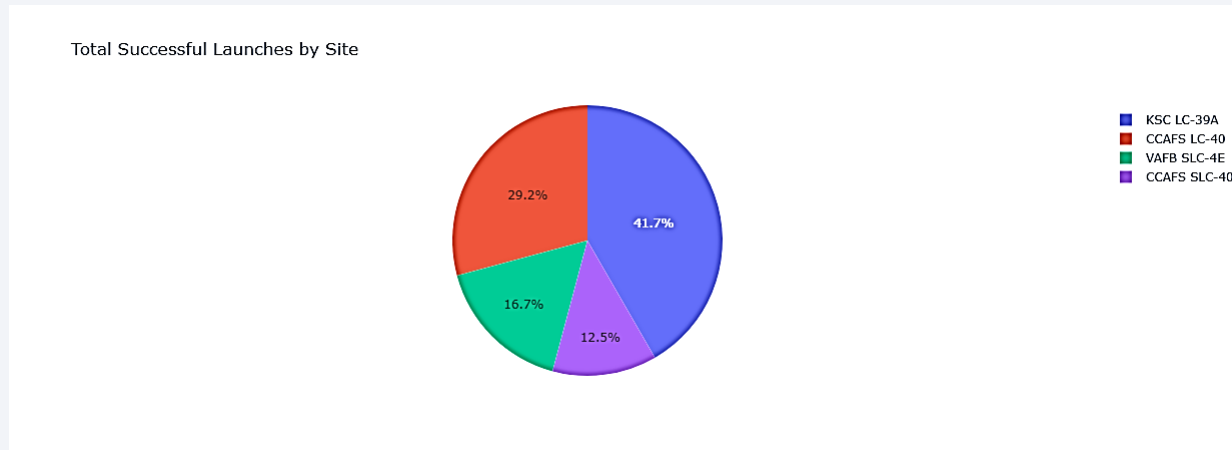
- **Launch Site:** Space Launch Complex 40 (Cape Canaveral Space Force Station)
- **Closest Railway:** Florida East Coast Railway (FEC) near Port Saint John
- **Straight-Line Distance:** ~11.2 km (6.96 miles)
- **Orientation:** Railway lies **west** of SLC-40, across Indian River and Merritt Island
- **Rail Access Point:** Near **US-1 and FL-528 interchange**, supporting freight and industrial logistics
- **Strategic Relevance:**
 - Enables **heavy cargo transport** for launch infrastructure
 - Supports **multi-modal logistics** with proximity to highways and Space Coast Regional Airport



Section 4

Build a Dashboard with Plotly Dash

Launch Success Distribution by Site – SpaceX Dashboard Screenshot



Overview:

The pie chart titled "**Total Successful Launches by Site**" displays the proportion of successful SpaceX launches across four major launch sites. The dropdown above the chart allows filtering by site, but in this view, "**All Sites**" is selected.

Key Elements:

•Launch Sites Represented:

- **KSC LC-39A** (Blue): 41.7%
- **CCAFS LC-40** (Red): 29.2%
- **VAFB SLC-4E** (Green): 16.7%
- **CCAFS SLC-40** (Purple): 12.5%

•**Chart Type:** Pie chart visualizing the relative success count per site.

•**Dropdown Filter:** Enables selection of individual sites to view detailed success vs. failure breakdowns.

Insights:

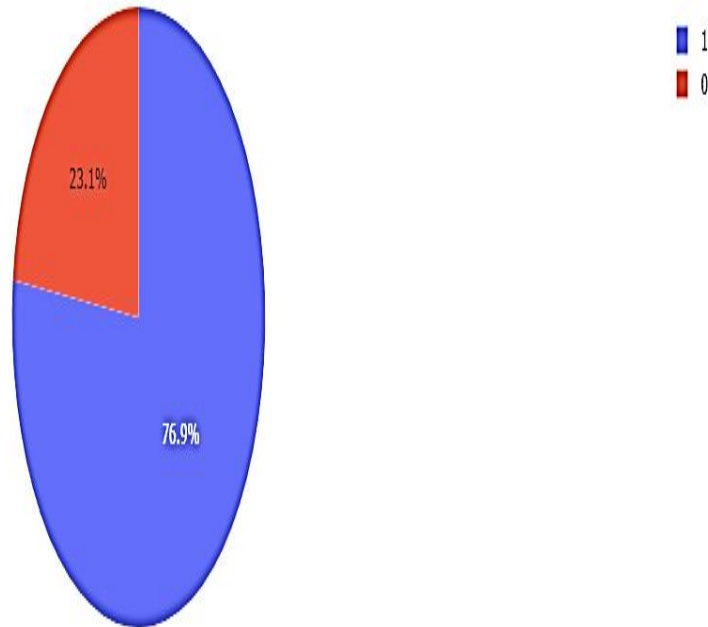
•**KSC LC-39A** leads with the highest share of successful launches, indicating it may be the most utilized or reliable site.

•**CCAFS LC-40** also contributes significantly, while **VAFB SLC-4E** and **CCAFS SLC-40** have smaller shares.

•This distribution helps identify operational focus and performance across SpaceX's launch infrastructure.

Launch Performance at KSC LC-39A – Highest Success Ratio Site

Success vs Failure for site KSC LC-39A



The pie chart titled "**Success vs Failure for site KSC LC-39A**" displays the launch outcome distribution for SpaceX missions from this site. The dropdown above the chart filters by launch site, and **KSC LC-39A** is selected — the site with the highest success ratio.

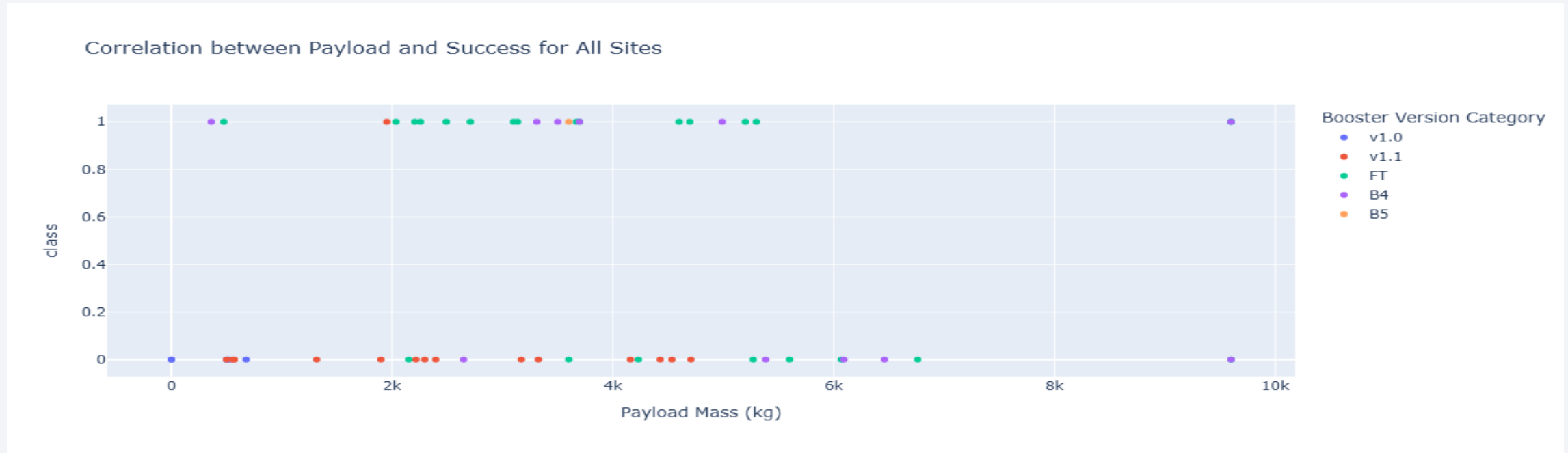
Key Elements:

- **Launch Site Selected:** KSC LC-39A
- **Success Rate:** 76.9% (blue segment, class = 1)
- **Failure Rate:** 23.1% (red segment, class = 0)
- **Legend:** Blue = Success, Red = Failure

Findings:

- KSC LC-39A demonstrates the **highest reliability** among SpaceX launch sites.
- The **high success ratio** suggests strong operational performance and possibly newer booster versions or optimized launch conditions.
- This site is a **strategic asset** for SpaceX's mission planning and reliability metrics.

Payload vs. Launch Outcome – Booster Performance Across All Sites



The scatter plot titled "**Correlation between Payload and Success for All Sites**" displays launch outcomes (success = 1, failure = 0) against payload mass. Each point is color-coded by **Booster Version Category** (e.g., v1.0, v1.1, FT, B4, B5). The **payload range slider** above the chart allows filtering to observe trends across different payload bands.

Key Elements:

- X-axis:** Payload Mass (kg)
- Y-axis:** Launch Outcome (class)
- Color:** Booster Version Category
- Interactive Filter:** Payload range slider (0–10,000 kg)

Findings:

- FT and B5 boosters** show the highest concentration of successful launches (class = 1), especially in the **2,000–6,000 kg** payload range.
- v1.0 and v1.1 boosters** are more scattered and include more failures, particularly at lower payloads.
- Higher payloads (>6,000 kg)** are mostly handled by newer boosters (B4, B5), with strong success rates.
- The visualization suggests a **positive correlation between booster evolution and reliability**, especially for mid-to-heavy payloads.

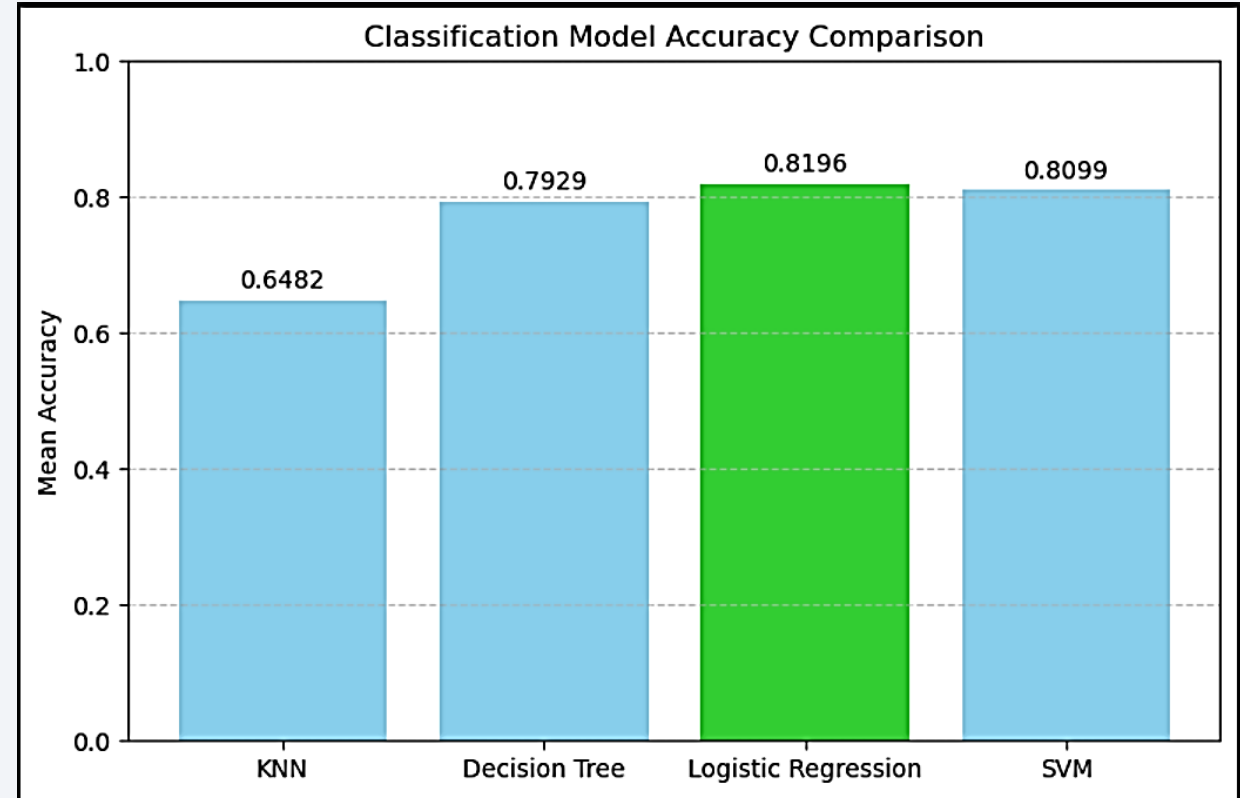
Section 5

Predictive Analysis (Classification)

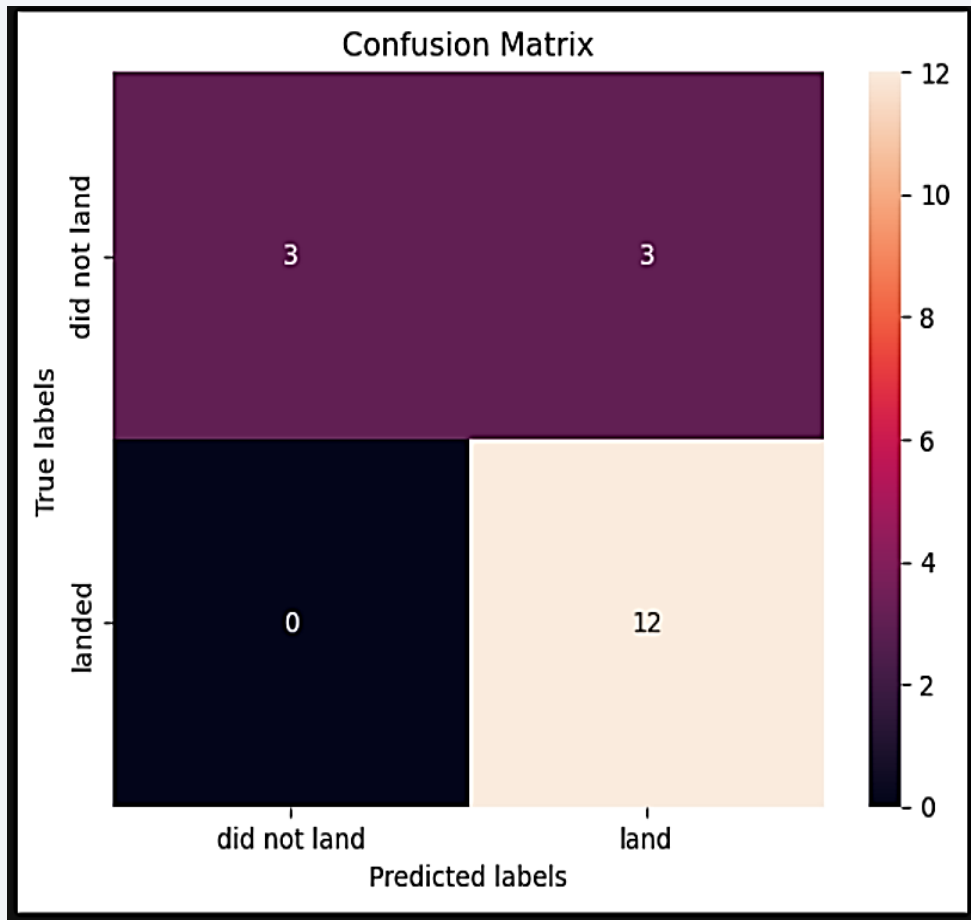
Classification Accuracy

Logistic Regression is best:

- **Highest Accuracy:** Outperforms all other models in mean accuracy.
- **Generalization:** Logistic Regression tends to generalize better than Decision Trees, which can overfit, and KNN, which struggles with noisy or high-dimensional data.
- **Balanced Performance:** While SVM is close (0.8099), Logistic Regression edges ahead, making it the most reliable choice here.



Confusion Matrix



- Axes:**

- X-axis:** Predicted Labels
- Y-axis:** True Labels
- Classes:** "did not land" vs "landed"

- Matrix Breakdown:**

- ☒ **True Positives** (landed & predicted landed): 12
- ☒ **False Positives** (did not land but predicted landed): 3
- ☒ **True Negatives** (did not land & predicted did not land): 3
- ☒ **False Negatives** (landed but predicted did not land): 0

- Key Insight:**

- The model shows **perfect recall** for successful landings (no false negatives)
- Slight overprediction of landings (3 false positives), but overall **strong classification performance**

- Use Case:**

Ideal for scenarios where **missing a successful landing is costly**, and **false alarms are tolerable**

Conclusions

- **Data Integration:** Combined SpaceX API and Wikipedia scraping to build a unified, clean Falcon 9 launch dataset.
- **EDA Insights:** Found that KSC LC-39A had the highest success rate; ES-L1, GEO, and SSO orbits showed near-perfect reliability.
- **Interactive Visuals:** Folium maps and Dash dashboard revealed strategic launch site logistics and booster performance patterns.
- **Modeling Outcome:** Logistic Regression emerged as the best classifier (CV: 0.8196, Test: 0.8333), with perfect recall for successful landings.
- **Operational Value:** The pipeline supports predictive planning, site evaluation, and booster reliability analysis for future launch strategies.

Appendix

- **Python Code Snippets**

→ API querying (requests), web scraping (BeautifulSoup, re, unicodedata), data wrangling (pandas), EDA (seaborn, matplotlib), classification modeling (sklearn)

- **SQL Queries**

→ Launch site filtering, payload aggregation, booster performance, landing outcome ranking, seasonal failure analysis

- **Charts & Visuals**

→ Scatter plots, bar charts, pie charts, confusion matrices, Folium maps, Dash dashboard components

- **Notebook Outputs**

→ Cleaned datasets: dataset_part_1.csv, spacex_web_scraped.csv, dataset_part_2.csv

→ Model tuning results and evaluation metrics

- **GitHub Repositories**

→ All notebooks and scripts available at: <https://github.com/ADVAIT135/IBM-DATA-SCIENCE/tree/4432ff75f43864b5185569860272509afa502183/10.%20Applied%20Data%20Science%20Capstone/Final%20Project%20Submission>

Thank you!

